Conflict, Security & Development

ISSN: 1467-8802 (Print) 1478-1174 (Online) Journal homepage: www.tandfonline.com/journals/ccsd20

Empirical evidence of direct and indirect relations between environmental pressure and conflict

Elizaveta Kopacheva, Giangiacomo Bravo & Davide Natalini

To cite this article: Elizaveta Kopacheva, Giangiacomo Bravo & Davide Natalini (24 Feb 2026): Empirical evidence of direct and indirect relations between environmental pressure and conflict, Conflict, Security & Development, DOI: 10.1080/14678802.2026.2625718

To link to this article: https://doi.org/10.1080/14678802.2026.2625718

Published online: 24 Feb 2026.

CONFLICT, SECURITY & DEVELOPMENT
https://doi.org/10.1080/14678802.2026.2625718

Empirical evidence of direct and indirect relations between environmental pressure and conflict

Elizaveta Kopacheva [a], Giangiacomo Bravo [b] and Davide Natalini [c]

[a]Department of Computer Science and Media Technology and Center for Data Intensive Sciences and Applications, Linnaeus University, Växjö, Sweden; [b]Department of Social Studies and Center for Data Intensive Sciences and Applications, Linnaeus University, Växjö, Sweden; [c]Global Sustainability Institute, Anglia Ruskin University, Cambridge, UK

ABSTRACT

At a time of increased negative effects of environmental change on human systems such as extreme meteorological events, resource scarcity, and refugee crises, it is especially pressing to understand what role environmental pressure plays in increasing the likelihood of violent conflict. We use a comprehensive measure of environmental pressure – *Population Biodensity*, defined as the ratio of a country's population to its biocapacity – to study the presence and intensity of conflict in a large dataset including 28 years of of observations for 181 countries, while controlling for the mediating effects of political regime and income. By means of Bayesian structural equation modelling, we found evidence of a significant relation between environmental pressure and both conflict presence and intensity. Moreover, environmental pressure was shown to be the most significant variable influencing the number of casualties in intrastate conflicts when compared to the level of democratisation and national income. These results highlight the importance of the natural environment for social sustainability, peace, and prosperity.

KEYWORDS

Environmental pressure; resource scarcity; Population Biodensity; intrastate conflict; Bayesian Structural Equation Modelling

Introduction

The link between natural resources and conflict has led to a heated debate in the literature at the crossroad between social and environmental sciences.[1] A recent metaanalysis concluded that, based on different mechanisms and under specific conditions, both the abundance of non-renewable, high profitable resources and the scarcity of renewable ones increase the risk of armed intrastate conflicts.[2] While this conclusion is supported by other literature in the field and the paths leading from resources to conflict have been detailed in several works,[3] the number and scope of empirical indicators testing these paths are limited. As a result, evaluations tend to be partial, focusing on specific and somewhat arbitrarily-selected resources or environmental issues.[4]

CONTACT Elizaveta Kopacheva elizaveta.kopacheva@lnu.se

What is missing in the current debate is a deeper understanding of whether the relation between natural resources and conflict also holds when taking into account the broader pressure of human consumption on the natural environment, e.g. as measured by aggregate indicators of environmental impact such as the *Ecological Footprint*.[5] In other words, our goal is to contribute to the existing literature by studying under which conditions environmental pressure increases the probability and intensity of conflict in countries, while being aware that this is not the only relevant factor and that its role is mediated by other variables, such as the quality of governance or the presence of ethnic divides.[6]

Although environmental pressure is not the only relevant variable, the growth of the human population and economy contributed to the overshoot of several planetary boundaries,[7] which in turn may affect conflictuality. This paper aims at contributing to the renewable resource-conflict debate by providing quantitative evidence linking environmental pressure and the presence and intensity of conflict in countries, whilst controlling for national, social and political conditions that exacerbate these relations. To do this, we analyse the relation between a novel indicator of environmental pressure called *Population Biodensity*[8] taking into account the mediation of political and economic factors[9]– specifically, political regime and income – with further factors included as robustness tests in the Appendix.

Population Biodensity (hereafter *PB*) is defined as the ratio of human population to biocapacity of a given area, where biocapacity is an indicator of biological productivity developed by the Global Footprint Network to estimate the number of ecological services provided by natural and man-managed lands.[10] It is intuitively linked to the idea of population density but, rather than simply counting the number of people living in a given area, it estimates the pressure they exert on the natural systems in which they live, mainly for the production of food and fibres and for the capture of their carbon emissions.[11] To put it simply, a higher *PB* means, all else being equal, an increased burden on the ecosystem due to the increased size of the population using those resources.

Although population density indicators have been used before to predict conflict,[12] the evidence for their effect is controversial and possibly mediated by other factors, such as institutions, ethnic diversity and income.[13] While population density simply measures the number of people per unit of area, *PB* contrasts the population size with the biological capacity of the land, providing a more nuanced view of human pressure on the environment. This indicator allows for the evaluation of whether a region's population is living within the ecological limits of its environment, rather than just quantifying crowding. This is a key contribution of this paper, as *PB* directly links human presence to ecological sustainability allowing a more comprehensive investigation of the effect of environmental pressure on conflict.

To the best of our knowledge, this is the first time that a comprehensive measure of environmental pressure is used to analyse conflict. In addition, and unlike previous studies using binary measures of conflict,[14] we focus on both the presence and intensity of intrastate conflict, the former measured as whether a country is experiencing any conflict in a year, and the latter measured in terms of the number of causalities per year (see Data and methods). The key novel contribution to knowledge of this paper lies in the comprehensiveness of our analysis. The main hypothesis we test in the next sections is that higher *PB* leads, all else being equal, to an increased likelihood and intensity of intrastate conflicts (see Previous works on factors affecting intrastate conflicts). Overall, we found evidence supporting this hypothesis, highlighting the importance of the environmental dimension and sustainable use of resources for human peace and prosperity.

In the next section, we will first introduce the background and literature on the relation between resources, conflict, income, and democracy and then introduce our model. In Data and methods we will present the methods and data used to test our hypotheses and we will present the results in Results. Furthermore, we will discuss the results in Discussion and underline the conclusions in Conclusions.

Research background and hypotheses

Previous works on factors affecting intrastate conflicts

Environmental pressure and conflict

With the declining number of interstate conflict,[15] recent conflict studies have increasingly focused on contemporary intrastate conflicts, which tend to be characterised by higher levels of violence compared to similar ones in the past.[16] Reviews and metaanalyses in the field of environmental security tend to agree on the existence of a relation between various environmental measures and conflict.[17] For instance, in their synthesis of the literature, Hsiang Burke Miguel (2013) concluded that, for each standard-deviation change in climate towards warmer temperatures or more extreme rainfall, there is a 4 per cent increase in the frequency of interpersonal violence and a 14 per cent increase in intergroup conflict.[18] In another recent review, Sweijs et al. categorised the different mechanisms linking climate change to violent conflict into ‘pathologies’,[19] whereby climate change indirectly affects conflict through other dynamics such as resource scarcity, migration, by generating social unrest, and finally through the implementation of policies to counteract climate change (which can have adverse effects to specific groups). Cappelli et al. also found that climate change is a threat multiplier,[20] which indirectly increases the risk of conflict in the case of pre-existing local vulnerabilities. Recent research similarly emphasises the role of climate-related disasters and environmental shocks as threat multipliers that can escalate ongoing conflicts, especially in ethnically fractionalised or vulnerable societies.[21] Disasters such as droughts and earthquakes can increase competition for scarce resources, leading to higher conflict intensity and fatalities.[22] However, the effect is not uniform; in some cases, disasters can also reduce conflict intensity by weakening one or more parties.[23]

More generally, environmental change and resource scarcity have been recognised as a factor that increases the risk of violent conflict.[24] Some studies find that environmental pressure and scarcity of renewable resources are significant predictors of conflict and intensity, with direct links to the number of casualties in some contexts.[25] For instance, the scarcity of natural resources seems to be related to the severity of conflict, with harvest loss found to be related to higher levels of political violence.[26] This could be explained by the additional pressure that resource scarcity can put on people’s lives and livelihoods (for instance, those connected to agriculture), which can result in dynamics that can further exacerbate underlying ruptures, although the presence of a direct, robust link between resource scarcity and conflict intensity is still debated.[27]

Although our focus is on the effect of environmental pressure more than on the exploitation of natural resources, it is also worth mentioning the 'resource curse' hypothesis, which suggests instead that the presence of valuable, easily extractable resources (oil, diamonds, minerals) can fuel conflict by providing funding for armed groups and incentives for rebellion.[28] Empirical evidence shows that resource abundance is associated with increased risk and duration of conflict, particularly when resources are 'lootable' and located within conflict zones.[29] However, the effect on conflict intensity (deaths) is often mediated by governance, ownership structures and the type of resource.[30]

Although research broadly highlight the relation between resource scarcity and conflict, several factors can mediate it, including institutional actions, inequality and migration.[31]Several indirect effects also link scarcity to conflict. Resource scarcity was found to negatively influence economic performance[32] and governance quality[33] and to increase migration, which in turn may affect conflict in the receiving country[34] (see Factors mediating the relation between environmental pressure and intrastate conflict).

To sum up, we can conclude that there is an agreement on the relation between natural resource scarcity and intrastate conflict, with social, political and economic factors mediating this relation. We therefore hypothesise a positive relation between our measure of environmental pressure (i.e. *PB*) and the presence of conflict (see Conceptual model and hypotheses). Similarly, for conflict intensity we hypothesise a positive relation between PB, our measure of environmental pressure and the intensity of conflict.

A word of caution comes from a branch of the literature showing how, under specific conditions, scarcity can foster cooperation rather than conflict, especially when it is moderate, perceived as a shared challenge and managed through inclusive systems.[35] However, these results are often related to specific resources, such as water,[36] or come from more abstract laboratory or simulation studies[37] and may not extend to more comprehensive measures of environmental pressure that consider a number of resources or real-world settings.

To complicate things further, the relation between availability of resources and conflict may be bidirectional - i.e. conflict may also affect the amount and availability of natural resources. Several factors possibly mediate this effect,[38] possibly with feedback loops between these variables. However, the specific method used in this paper (see Data and methods) and the robustness tests presented in the Appendix allow us to test for a specific causality direction,[39] which we hypothesise goes from environmental pressure to conflict, with the additional analyses presented in the Appendix (Figures A1 and A2) further strengthening our arguments.

Factors mediating the relation between environmental pressure and intrastate conflict

As mentioned in the previous section, research suggests that the relation between resource scarcity and conflict is mediated by a number of social-economic factors. A prominent one is the level of income, usually measured as per capita Gross Domestic Product (GDP). Several studies indeed found a link between low economic development and conflict.[40] For instance, Fearon and Laitin[41] and Collier and Hoeffler[42] found that poverty, low economic growth and dependency on primary commodity exports are associated with both inter and intrastate conflicts. Vice-versa, higher income allow to reduce local environmental pressure and import crucial resources from other countries via international trade - something often called displacement of environmental costs[43]– which, in turn, is likely to reduce the local occurrence of conflict.[44] Based on these considerations, we expect a negative association between a country's income and conflict (see Conceptual model and hypotheses).

Beyond income, a country's institutional and political context critically shapes the relation between environmental pressure and conflict. The effectiveness, inclusiveness, and legitimacy of political institutions determine whether resource-related grievances are managed through peaceful mechanisms or escalate into violence.

Research examining this dimension generally agrees with the democratic civil peace hypothesis, which holds that democracies are less prone to intrastate conflict because they institutionalise political participation, allow peaceful dissent and are less inclined to resort to violence in addressing internal disputes.[45] By contrast, non-democratic regimes are generally more prone to conflict – with corruption and repression often acting as triggers – and more likely to respond violently to initially non-violent dissent.[46]

The evidence becomes more nuanced when examining democratisation on a spectrum. Early studies suggested an inverted-U relation, in which hybrid regimes – combining democratic and authoritarian features – are most prone to conflict due to institutional instability and contested legitimacy.[47] More recent research challenges this view, finding that only consolidated democracies consistently experience lower conflict risk, while hybrid regimes and autocracies are similarly likely to experience conflict.[48]

In general, while institutional design and the management of group grievances are central to how conflicts emerge and escalate, there seem to be many mechanisms at play.

Redistribution and equality, more commonly found in democracies, are generally associated with lower risks of violent conflict.[49] Conversely, high and politically salient inequalities – particularly when aligned with ethnic, religious, or regional divisions and reinforced by political exclusion – serve as powerful drivers of group mobilisation and violent conflict.[50]

Property regimes and land tenure are key characteristics that differentiate different types of regimes and represent another critical factor in shaping conflict risk. Empirical studies show that insecure tenure, overlapping claims and forced displacement create mobilisable grievances, especially where rural livelihoods depend heavily on land access.[51]

Given the variety of mechanisms at play, findings on the relation between regime type and intrastate conflict remain mixed. Nevertheless, one pattern is consistently observed: consolidated democracies experience significantly fewer intrastate conflicts than other regime types. Based on this evidence, we expect a negative correlation between democratic regimes and conflict.

Conceptual model and hypotheses

The main hypotheses of this work are that countries with higher levels of environmental pressures tend to experience more frequent intrastate conflicts (H1), as well as higher intensity of conflict, leading to more casualties (H2). This outcome, however, is expected to be mediated by national income and the political regime.

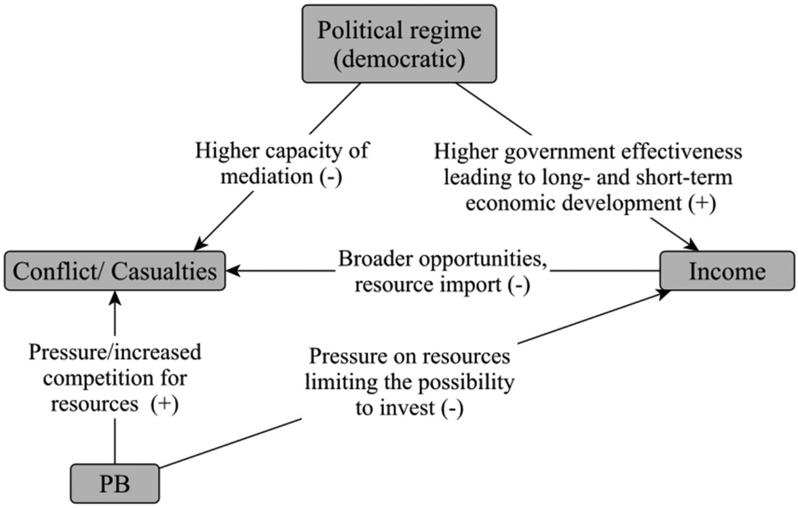

Figure 1. Conceptual model of the relations among variables. Labels on each path summarize the expected mechanisms affecting the relation. A (+) symbol indicates that a direct relation is expected, a (–) symbol indicates an inverse expected relation.

In addition, we expect to see a negative effect of *PB* on GDP as suggested by several works.[52] These studies examined the link between rising temperatures and reduced precipitation and the decrease of agricultural yields, thus, a negative impact on GDP. In addition, GDP is experted to exert a direct negative effect on conflict, as suggested by other studies.[53] Furthermore, we expect democratic development to positively affect GDP.[54]

Based on these considerations, Figure 1 provides a conceptual overview of the expected relations between the variables of interest for this research and sketches the structure of the models that will be tested in Results.

Data and methods

While national-level analysis has been common in studies of environmental conflict, some of the literature quoted above suggests that focusing on the subnational level may provide a more nuanced and accurate understanding of conflict dynamics.[55] Nevertheless, large-N national-level studies remain valuable for understanding broader trends and policy impacts[56] and, crucially, most of the data are only available on a sufficient spatial and temporal scale only at this level. This is especially true for *biocapacity* data, which are typically computed at the national level and represent the crucial input to derive our *PB* indicator.[57] Although acknowledging this as a possible limitation of the current approach, this choice allows us to test our model with a much larger and more comprehensive dataset that would otherwise have been possible.

Dataset

The data used in the analysis includes 181 countries and refer to the 1992–2019 period, 2019 being the last year with available biocapacity data.[58] We excluded countries with less than 5 observations (i.e. less than 5 years of data) within the time period considered in the analysis. Following Tamburino and Bravo,[59] countries with a surface area < 10000 km^2 — which are often city-states presenting extreme *PB* values - were also excluded from the analysis.

The data is structured as an annual country-level panel. Data for conflict events and casualties were extracted from the Uppsala Conflict Data Program (UCDP) database,[60] data on countries' political regimes were from the V-Dem political regime index,[61] data on countries' GDP were extracted from the World Bank Open Data[62] and *PB* data were calculated for 2019 following the procedure in.[63]

To obtain a comprehensive picture of the overall number of intrastate conflict events in countries, we merged the UCDP 'intrastate' and 'intrastate with foreign involvement' conflict data in a single dataset. Intrastate conflicts are defined as conflicts between the state and one or several rebel groups.[64] Intrastate conflicts with foreign involvement imply the involvement of foreign governments with troops. In both types of conflict, insurgents pursue territorial secession or autonomy and both are a measure of the state's internal tensions.[65] We acknowledge that there is a multitude of types of intra-state conflict which differ in terms of micro-level and contextual mechanisms. However, they share key underlying dynamics, such as those highlighted in the previous section, that are of interest to the analysis of this paper.

The outcome variable (i.e. the terminal node in Figure 1) used in the analysis is the UCDP's best estimate for battle-related deaths. This variable is used both as it is (hereafter referred to as *Casualties*) and transformed into a binary *Conflict* variable, taking the value of 1 if the country had at least one intrastate conflict in a given year and 0 otherwise. In line with the UCDP's definition of an armed conflict, we consider a country to be part of a conflict if this leads to at least 25 violent deaths per year. The *Conflict* variable is used to estimate the probability that a country enters a conflict, while the *Casualties* one is used as a proxy for conflict intensity. How these two variables are used in our models is detailed in Analysis.

The main independent variable in our model, the *Population Biodensity* (*PB*), is defined as $PB = P/BC$, where P is the country's population and BC its total biocapacity and is measured in people/global hectares (gha).[66] A country's *BC* is defined as 'the ecosystems' capacity to produce biological materials used by people and to absorb waste material generated by humans, under current management schemes and extraction technologies'.[67] *P* data were extracted from the World Bank Open Data,[68] while *BC* ones derive from the Global Footprint Network (GFN) National footprint accounts.[69] Assuming constant consumption, a higher value of the PB indicator represents a situation where a large population put a heavy weight on local ecosystems. On the contrary, smaller values of the indicator represent a more sustainable situation where resources are abundant compared to the population that relies on them.

The use of the *PB* indicator as a predictor for intrastate conflict is supported by previous literature suggesting that environmental stress accumulates over time until it causes a sudden explosion of violence.[70] *PB* indeed incorporates the current (as population pressure) and accumulated stress (as reduced biocapacity) on natural systems, making it apt to explain events that may have roots extending into the past.

We accounted for the influence of income and political regime of a given country by including in the models the *per capita* GDP, estimated in purchasing power parity (PPP) terms and the V-Dem political regime index. V-Dem identifies the political regime based on the competitiveness of access to power and liberal principles. The political regime index is an aggregation of several V-Dem indexes such as the equality in access to justice, fairness of elections and transparency of law enforcement.[71] We used the V-Dem political regime index to build a *Political regime* variable including 4 ordered categories: closed autocracies, electoral autocracies, electoral democracies and liberal democracies.[72]

A methodological note on the conflict measure used in this paper: our analysis focuses on the *presence* of conflict rather than its *onset*. In our dataset, the dependent variable 'conflict' is binary, taking the value 0 to indicate the absence of conflict in a given country-year and 1 to indicate its presence. This differs from measures of conflict onset, where a value of 1 would denote the initial occurrence of a conflict event (onset) within the observed period. Because our unit of analysis is the country-year, a country can have a value of 1 for multiple consecutive years if the same conflict persists - hence, our variable captures conflict presence, not onset. This choice is primarily methodological: given that our dataset spans a maximum of 28 years, focusing on onset would yield a large number of zeros and very few ones, thereby reducing the statistical power of our analysis. As for our definition of intensity, we don't capture the *overall* intensity of a conflict, as our dependent variable is not cumulative. In our analysis, with conflict intensity we capture *number* of deaths per year in the country, which means that they may not be necessarily related to one single conflict. Similarly, the rationale for this choice is methodological, to reduce the probability of autocorrelation in the dependent variable.

Analysis

To test our hypothesis, we used a two-step analytical strategy based on some of the advanced methods advocated by the previous literature to better understand the causal pathways linking environmental pressure with conflict.[73] First, we estimated a Bayesian structural equation model (SEM) having the structure shown in Figure 1 and including a logistic regression for the model using the *Conflict* variable as terminal node (The conflict model). This model is designed to test the hypothesis that countries with higher *PB* have a higher probability of experiencing violent intrastate conflict (H1). The model is defined as follows:

$$\text{Income}_{ij} = \beta_0 + \beta_1 \times \text{PB}_{ij} + \beta_2 \times \text{Political regime}_{ij} + u_j^{(1)} + \varepsilon_{ij}^{(2)} \tag{1}$$

$$\text{Conflict}_{ij} = \alpha_0 + \alpha_1 \times \text{PB}_{ij} + \alpha_2 \times \text{Political regime}_{ij} + \alpha_3 \times \text{Income}_{ij} + u_j^{(2)} + \varepsilon_{ij}^{(2)} \tag{2}$$

where Income_{ij}, PB_{ij} and $\text{Political regime}_{ij}$ are the income level, PB and political regime of country j in year i, respectively; $u_j^{(1)}$ and $u_j^{(2)}$ are random effects for each country; and $\varepsilon_{ij}^{(1)}$ and $\varepsilon_{ij}^{(2)}$ are the error terms. Note that the country variable has been included in the models as a random-effect (random intercept). This is due to the variability of unobserved country-specific characteristics over the examined time-period. Additionally, including country as a random-effect enables generalisation. Estimating random effects controls for variance across countries, allowing conclusions about the entire population and compensating for missing cases.

The second model was then estimated to test the hypothesis that *PB* also affects the intensity of conflict (H2). The new model has the same structure as the previous one but uses the *Casualties* variable as terminal node (The casualties model). Moreover, the countries included in the model estimation are only the ones that actually experienced a conflict in a given year, i.e. the ones with *Conflict* = 1.

The advantage of this two-step approach is that we can use all the available data to test H1, while limiting our attention to countries that did actually experience conflicts when testing H2. As only a minority of countries were actually involved in conflicts leading to at least 25 violent deaths in a given year, using the whole dataset in the second model would have only inflated the number of zeroes (i.e. no casualties) in the outcome variable. Focusing only on countries that actually experienced conflicts allowed us to circumvent this undesirable outcome distribution, hence preventing potential biases in the model estimation.

We used a Bayesian approach in all our estimations because it allowed greater flexibility in defining the SEM structure and minimised the risk of biases due to the violation of assumptions of the maximum likelihood estimator.[74] A random effect (RE) for countries was included in all regressions, which allowed accounting for repeated observations of the same country. In addition, regressions having the binary variable *Conflict* as outcome had a logit specification. All numerical variables were standardised prior to model estimation, while the *Political regime* one was included using closed autocracies as reference level.

Additionally, we conducted a series of robustness tests to ensure the reliability of our results. The outcomes of these tests are detailed in the Appendix. Firstly, we estimated both models after normalising the numeric variables using log-transformations (see Tables A1 and A2). Secondly, we examined the delayed effects of PB on the outcome variables, namely conflict presence and intensity (see Tables A3–A6). This involved analysing the impact of PB at time $t-1$ and $t-2$ on conflict at time t.

Next, we performed a robustness test to investigate how *PB* influences conflict within subsets of countries characterised by varying levels of natural resource rents (see Tables A7–A10). Furthermore, we evaluated our measure of environmental pressure against variables from competing explanations, such as cultural diversity (Tables A11 and A12), immigration (Table A13), inequality (Tables A14–A19) and property rights (Tables A20 and A21). Lastly, we applied Bayesian network structure learning to examine whether our assumptions about causal directionality were supported by the data (see Figure A1 and A2). After using different algorithms (hill-climbing and tabu search) and multiple scoring criteria (BIC, AIC and BDe), we found that the learned network structures closely matched our hypothesised model. Overall, across all robustness checks, the results consistently supported our original findings, demonstrating that *PB* is a significant predictor of both conflict presence and intensity.

All models were estimated on the *R 4.3.2* platform[75] using the *brms* package[76] (the R code is available upon request).

Results

The conflict model

Figure 2 shows countries that experienced at least one conflict in the 1992–2019 period. Over 40 per cent of the world's countries fall in this category, with the majority of them located in Africa and Asia.

A simple test of the hypothesis that countries experiencing conflicts have higher *PB* than peaceful ones[77] showed promising evidence of the link between environmental pressure (as measured with this indicator) and the presence of conflict (r =0.707, BF = 35:1).[78] However, this statistic does not take into account the effect of the different covariates discussed above and may be biased because of repeated observations for each country.

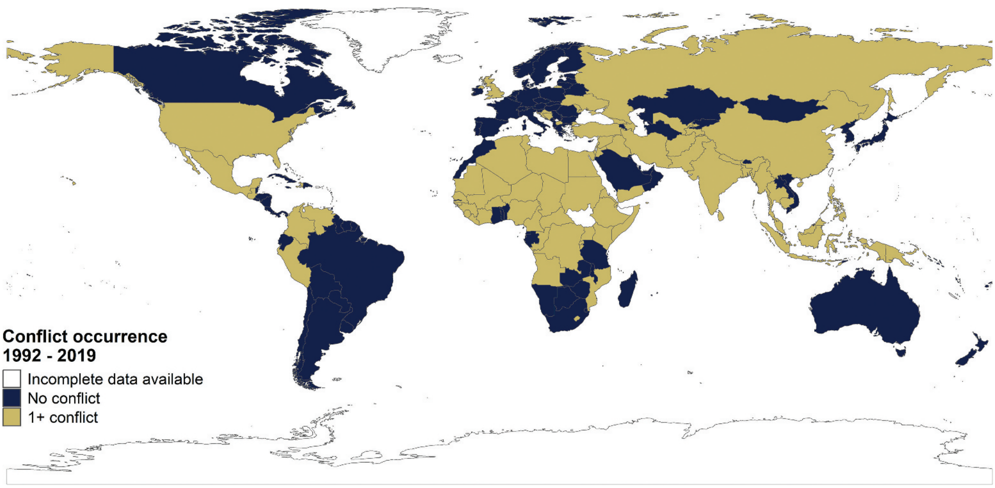

Figure 2. Presence of conflict in countries during the 1992–2019 period.

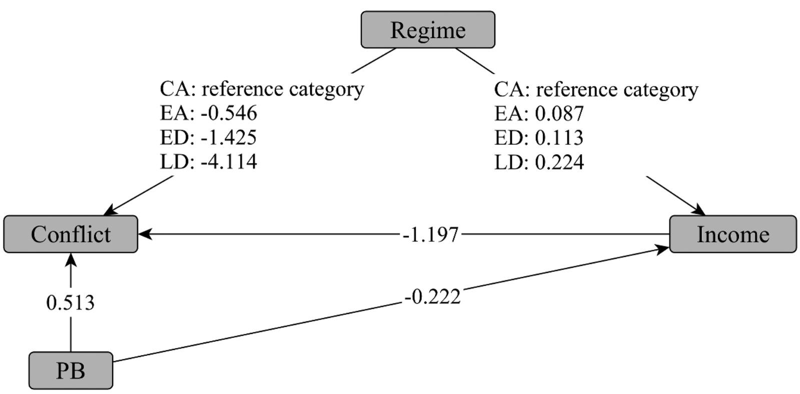

Figure 3. Conflict model. Paths represented by plain arrows have a BF ≥ 10, indicating substantial evidence for the hypothesized relation, path with dashed arrows have a BF < 10. Legend: CA, closed autocracies; EA, electoral autocracies; ED, electoral democracies; LD, liberal democracies.

The estimated Bayesian SEM designed to take these issues into account is shown in Figure 3, while Table 1 reports the details of coefficient estimates. All paths show effects consistent with the ones hypothesised in Figure 1. Specifically, our hypothesis of a positive relation between *PB* and conflict (H1) is supported by the data. There is substantial-to-strong evidence for both a direct effect of *PB* on conflicts and an indirect one mediated by income (Table 1). The total (direct + indirect) effect of *PB* on conflict hence is 0.779, even stronger than what resulted from the simple analysis presented above.

The model also highlighted strong evidence of a (direct + indirect) effect of the political regime on conflict, with democracies less likely to experience deadly intrastate conflicts. Moreover, the more a country approximates the ideal of liberal democracy the lower the probability of deadly intrastate conflict. Taking into account both direct and indirect effects, the odd ratios of conflict onset when increasing *PB* by one standard deviation are 2.18 (i.e. more than double). The odds of conflict onset for electoral democracies compared to the reference category (closed autocracies) are 0.21 of the ones for closed autocracies. This already large gap increases further for liberal democracies, with the odd ratio becoming 0.013. This extremely low value is explained by the fact that, out of 44 countries classified as liberal democracies in the period under consideration, only three did experience intrastate conflicts, namely the UK (with one conflict year in the 1992–2019 period), USA (with 18 out of 28 conflict years and 5513 casualties) and Israel (with 24 out of 28 conflict years and 8280 casualties). Finally, an increase by one standard deviation of per capita GDP directly reduces conflicts, with an odd ratio = 0.30 (Table 1).

The casualties model

To test H2, we estimated a second Bayesian SEM with the number of casualties as the final node and only including the countries actually involved in at least one deadly intrastate conflict in a given year (i.e. 74 countries in total).

Figure 4 shows the total casualties per country between 1992 and 2019. Only 2 out of 181 countries with available data in the period under consideration experienced conflicts involving more than 100,000 casualties. These countries are Syria, with 9 out of 28 conflict years and 327,426 casualties and Afghanistan with 28 out of 28 conflict years and 229,449 casualties. In addition, the list of the top 15 countries with the highest number of casualties includes Iraq, Sri Lanka, Bosnia and Herzegovina, Turkey, India, Pakistan, Yemen, Somalia, Angola, the Democratic Republic of the Congo, Colombia, Russia and Nigeria.

Table 1. Coefficient estimates for the conflict model.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	−0.122	[−0.282, 0.04]	1:14
Income	PB	−0.222	[−0.245, −0.198]	1:100000
Income	Regime: Electoral autocracies	0.087	[0.05, 0.123]	100000:1
Income	Regime: Electoral democracies	0.113	[0.071, 0.154]	100000:1
Income	Regime: Liberal democracies	0.224	[0.161, 0.287]	100000:1
Conflict	(Intercept)	−3.466	[−4.556, −2.483]	1:100000
Conflict	PB	0.513	[−0.024, 1.055]	32:1
Conflict	Income	−1.197	[−2.047, −0.364]	1:416
Conflict	Regime: Electoral autocracies	−0.546	[−0.999, −0.095]	1:113
Conflict	Regime: Electoral democracies	−1.425	[−2.003, −0.852]	1:100000
Conflict	Regime: Liberal democracies	−4.114	[−6.999, −1.67]	1:4544

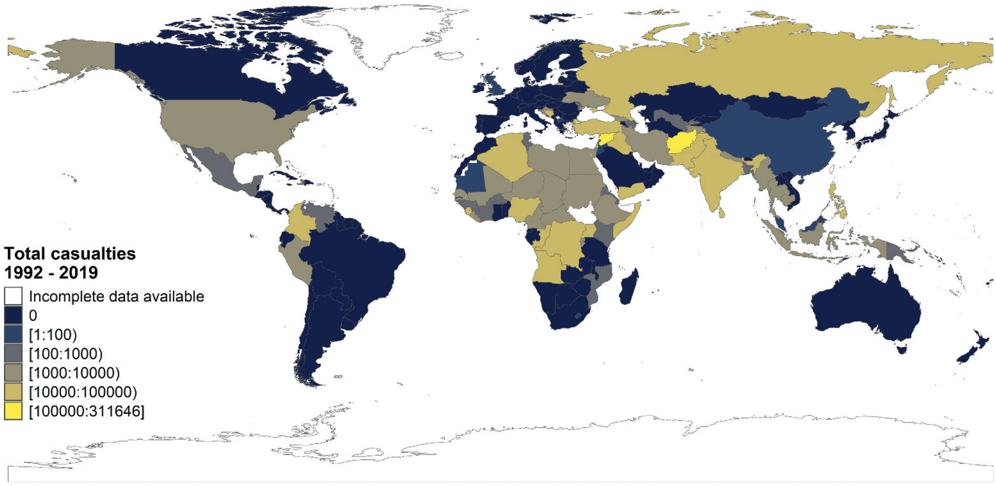

Figure 4. Casualties per country in the 1992–2019 period.

Figure 5 summarises the model results while Table 2 shows the details of parameter estimations. There is very strong evidence for the direct effect of *PB* on the intensity of conflict (i.e. casualties), which is fully consistent with H2. In contrast to the previous model, *PB* is the only variable showing substantial-to-strong evidence of an effect larger than zero. Moreover, the size of the effect of *PB* is remarkable, with an increase of one standard deviation in *PB* increasing the number of casualties by 1037 units.

National income, as expected, has a direct negative effect on the number of casualties. Specifically, a one standard deviation increase in national income reduces the number of casualties by 699 units.

Nonetheless, not all paths in the casualties model reflected the ones assumed in Figure 1. The main discrepancies were the lack of evidence of a direct effect of political regime on causalities and the fact that, despite our expectations, there is strong evidence of a positive effect of *PB* on income for countries experiencing conflict. These unexpected results will be discussed in Discussion.

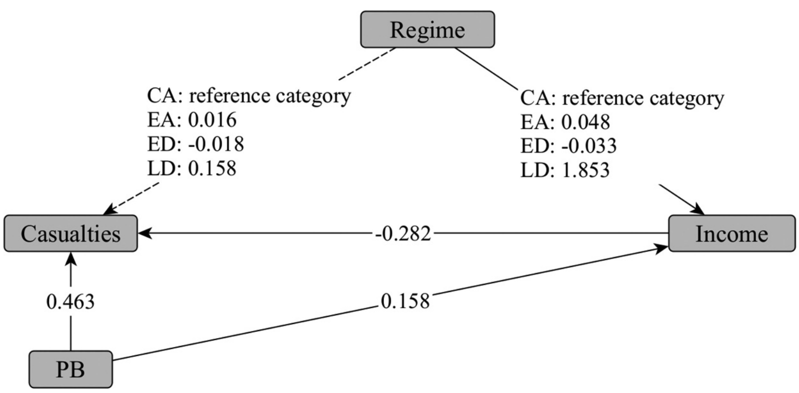

Figure 5. Casualties model. Paths represented by plain arrows have a BF ≥ 10, indicating substantial evidence for the effect, path with dashed arrows have a BF < 10. Legend: CA, closed autocracies; EA, electoral autocracies; ED, electoral democracies; LD, liberal democracies.

Table 2. Coefficient estimates for the casualties model.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	−0.62	[−0.702, −0.537]	1:100000
Income	PB	0.158	[0.098, 0.22]	100000:1
Income	Regime: Electoral autocracies	0.048	[0.019, 0.077]	1428:1
Income	Regime: Electoral democracies	−0.033	[−0.076, 0.01]	1:14
Income	Regime: Liberal democracies	1.853	[1.47, 2.231]	100000:1
Casualties	(Intercept)	0.063	[−0.279, 0.403]	2:1
Casualties	PB	0.463	[0.192, 0.744]	2325:1
Casualties	Income	−0.282	[−0.675, 0.103]	1:12
Casualties	Regime: Electoral autocracies	0.016	[−0.182, 0.212]	1:1
Casualties	Regime: Electoral democracies	−0.018	[−0.297, 0.262]	1:1
Casualties	Regime: Liberal democracies	0.158	[−0.96, 1.295]	2:1

Discussion

Our analysis showed that environmental pressure - estimated here using the *Population Biodensity* indicator[79]— affects the presence and intensity of deadly intrastate conflicts both directly and indirectly. Controlling for the influence of economic and political factors, measured here as per capita GDP (PPP) and political regime, we found that environmental pressure affects the presence of conflict and GDP in a country, which in turn has a significant negative relation with conflicts. In addition, all tests performed using subsets of countries or different indicators confirm the robustness of the relation between PB and conflict (see the Appendix).

These findings are consistent with earlier studies linking resource scarcity to conflict,[80] but extend this research in important ways. Most previous work focused on only one or two resources at a time - for example, food and water insecurity[81] or fuel scarcity.[82] By contrast, we used a comprehensive measure of environmental pressure. This allows us to conclude that environmental pressure as a whole (i.e. not focusing on any specific natural resources) plays a significant role in the dynamics that trigger intrastate deadly conflict.

Beyond the mere presence of conflict, our two-step analytical strategy provides novel evidence of the influence of environmental pressure on intrastate conflict intensity. We found that environmental pressure positively affects the number of casualties in intrastate conflicts. Furthermore, comparing the statistical effects of all the examined variables, we found environmental pressure to have the strongest relation to the number of intrastate conflict casualties. To our knowledge, this is the first study to demonstrate such a relation: earlier work has typically operationalised conflict outcomes as either the number of conflicts or as a binomial variable of conflict,[83] leaving intensity largely unexplored. This provides new evidence for a potentially different mechanism linking resources and conflict intensity, one not based on the need for resources to finance and sustain violence (i.e. the more resources available, the more the intensity of conflict)[84] but more in line with the mechanisms that rule the connection between scarcity of resources and conflict (e.g. the less resources available, the more the grievance of local populations, the more the intensity of conflict).

In short, our results highlight that environmental pressure contributes to intrastate conflict dynamics both directly and indirectly - through its impact on GDP - and amplifies violence within such conflicts, providing evidence of a larger effect than previously documented.

Turning to the influence of political and economic factors, we expected that democratic states would experience less frequent and less intense intrastate conflicts due to their capacity to accommodate grievances through non-violent channels.[85] This expectation is partly confirmed by our findings: we observe a negative association between democratic regimes and conflict. Moreover, the probability of conflict appears to decline steadily when moving from closed autocracies to electoral autocracies, electoral democracies and liberal democracies. These results challenge the early notion of an inverted-U relation between democracy and conflict[86] and instead align with research suggesting little differences between hybrid regimes and autocracies, with only consolidated democracies consistently showing lower risks of violence.[87]

Except for the political regime, previous research highlights a more nuanced set of mechanisms through which conflict emerges and escalates. The combination of governance quality and conditions, such as inequality – particularly when it occurs between different cultural groups or in contexts of high immigration[88]– and insecure land and property rights,[89] appears particularly important for explaining violent conflict. While these mechanisms were not the primary focus of our study, we conducted a series of robustness tests to assess the influence of PB relative to alternative explanatory factors.

The results show that, even after accounting for inequality, cultural diversity, immigration, and property rights, PB remains a significant predictor of both conflict presence and intensity (see Appendix).

Regarding unexpected findings, we observed a positive correlation between PB and GDP in the casualties model (see Figure 5 and Table 2). This likely reflects the relatively low variance of GDP within this model. As low levels of economic development and conflict are correlated[90]– a pattern also evident in our first model – most countries in the casualties model sample are lower-middle or low-income. These countries have little or no possibility to import large quantities of external resources to mitigate the pressure on local ecosystems. Consequently, the observed positive relation between PB and GDP may be an artefact of the of the (sub)sample including only few uppermiddle or high-income countries. Additionally, the limited degrees of freedom in the casualties model may explain the lack of a statistically significant effect of regime on casualties.

Other limitations of this study stem from the data used. Conflict events and casualties were drawn from the Uppsala Conflict Data Program database,[91] which defines intrastate conflicts as those with at least 25 deaths in a given year. Yet, previous studies[92] defined civil conflict as conflicts leading to at least one death per year. We assume that there may be missing observations in our dataset due to inconsistencies in the definitions.

Moreover, the data are measured at the country-year level, while intrastate conflicts often occur locally and environmental pressure can vary within a country. Therefore, our research design runs the risk of oversimplifying the relation between environmental pressure and the risk of conflict. We recognise this limitation and encourage future studies to address this gap.

Conclusions

In this study, we tested the significance of the relation between environmental pressure and intrastate conflict. We found that environmental pressure is positively related to conflict, both directly and indirectly through income as a mediator. Moreover, this statistical effect was stronger than the one for political regime and income for conflict presence and the only presenting very strong evidence of an effect for conflict intensity.

Future work in the field of environmental security should focus on developing more accurate and granular databases of intrastate conflicts. This will enable further testing of the relation between the 'health' (availability of resources, etc.) of the natural environment and conflict. Technological advancements could support these developments by harnessing the power of social media and citizen science, as demonstrated by projects like Crisis Mapping,[93] to collect real-time data via individuals witnessing the conflict.

Further research should also deepen understanding of the pathways and conditions through which environmental pressures translate into conflict. Our results indicate the existence of both direct and indirect links between resource scarcity and conflict presence and intensity, but more granular approaches are needed to specify these relations. This includes examining how governance quality, inequality, and property rights systems mediate or aggravate the effects of resource scarcity. Developing systematic approaches to identify and collect data on local property rights would be particularly valuable for enabling comprehensive, quantitative, and cross-national analyses of how different governance models influence conflict.

Notes

1. Abel et al., 'Climate, Conflict and Forced Migration'; Gargano et al., 'Migration and Conflict before Agriculture'; Kelley et al., 'Climate Change and the Syrian Drought'; Natalini et al., 'Political Fragility, Food Prices, and Food Riots'; Adams et al., 'Sampling Bias in Climate – Conflict Research'.
2. Vesco et al., 'Natural Resources and Conflict'.
3. Homer-Dixon, *Environment, Scarcity, and Violence*; Acemoglu et al., 'A Dynamic Theory of Resource Wars'; Nie, 'Drivers of Natural Resource-Based Political Conflict'.
4. Arezki et al., 'Oil Rents, Corruption, and State Stability'; Gleick, 'Water and Conflict'; Lagi et al., *Food Crises and Political Instability in MENA*.
5. Wackernagel et al., 'Ecological Footprint Accounts'.
6. Homer-Dixon, *Environment, Scarcity, and Violence*.
7. Steffen et al., 'Planetary Boundaries'.
8. Tamburino and Bravo, 'Reconciling Ecology and Human Development'.
9. Homer-Dixon, *Environment, Scarcity, and Violence*.
10. Global Footprint Network, 'Glossary'; Global Footprint Network, National Footprint Accounts.
11. Tamburino and Bravo, 'Reconciling Ecology and Human Development'; Global Footprint Network, 'Glossary'.
12. Fearon and Laitin, 'Ethnicity, Insurgency, and Civil War'.
13. Arbatlı et al., 'Diversity and Conflict'; Adanu, 'Population, Institutions, and Violent'.
14. Vesco et al., 'Natural Resources and Conflict'.
15. Oneal and Russet, 'The Classical Liberals were Right'; Peter Wallensteen, 'Universalism vs. Particularism'.
16. Snow, *Uncivil Wars*.
17. Bernauer et al., 'Environmental Changes and Violent Conflict'; Vesco et al., 'Natural Resources and Conflict'; Xie et al., 'The Impacts of Climate Change on Violent Conflict Risk'.
18. Hsiang et al., 'Quantifying the Influence of Climate on Human Conflict'.
19. Sweijs et al., *Unpacking the Climate Security Nexus*.
20. Cappelli et al., 'Local Vulnerabilities to Climate Change and Conflict in East Africa'.

21. Schleussner et al., 'Climate Disasters and Armed-Conflict Risk in Fractionalized States'; Brancati, 'Political Aftershocks'; Koubi and Spilker, *Natural Resources, Climate Change, and Conflict*; Ide, 'Rise or Recede?'
22. Ide, 'Rise or Recede?'; Schleussner et al., 'Climate Disasters and Armed-Conflict Risk in Fractionalized States'; Brancati, 'Political Aftershocks'.
23. Ide, 'Rise or Recede?'
24. Bernauer et al., 'Environmental Changes and Violent Conflict'.
25. Gawande et al., 'Renewable Natural Resource Shocks and Conflict Intensity'; Gizelis and Wooden, 'Water Resources, Institutions, & Intrastate Conflict'; McLaughlin Mitchell, 'Cross-Border Troubles?'; Schleussner et al., 'Climate Disasters and Armed-Conflict Risk in Fractionalized States'.
26. Wischnath and Buhaug, 'Rice or Riots'.
27. Koubi et al., 'Natural Resources and Armed Conflict'; Koubi and Spilker, *Natural Resources, Climate Change, and Conflict*; Vesco et al., 'Natural Resources and Conflict'; O'Brochta, 'A Meta-Analysis of Natural Resources and Conflict'; Bayramov, 'Review'; Mildner et al., 'Scarcity and Abundance Revisited'.
28. Conrad et al., 'Rebel Natural Resource Exploitation and Conflict Duration'; Rigterink, 'Diamonds, Rebel's and Farmer's Best Friend'; Chisadza et al., 'Giant Oil Discoveries and Conflicts'; Sorens, 'Mineral Production, Territory, and Ethnic Rebellion'.
29. Conrad et al., 'Rebel Natural Resource Exploitation and Conflict Duration'; Rigterink, 'Diamonds, Rebel's and Farmer's Best Friend'; Sorens, 'Mineral Production, Territory, and Ethnic Rebellion'.
30. Rigterink, 'Diamonds, Rebel's and Farmer's Best Friend'; Chisadza et al., 'Giant Oil Discoveries and Conflicts'; Denly et al., 'Do Natural Resources Really Cause Civil Conflict?'
31. Xie et al., 'The Impacts of Climate Change on Violent Conflict Risk'; Sweijs et al., *Unpacking the Climate Security Nexu*; van Baalen and Mobjork, 'Climate Change and Violent Conflict in East Africa'; Schleussner et al., 'Climate Disasters and Armed-Conflict Risk in Fractionalized States'.
32. Levy, 'Is the Environment a National Security Issue?'
33. Kahl, *States, Scarcity, and Civil Strife in the Developing World.*
34. Durham, *Scarcity and Survival in Central Americ*; Hawes, 'Theories of Peasant Revolution'; Reuveny, 'Climate Change-Induced Migration and Violent Conflict'.
35. Nie et al., 'Resource Scarcity and Cooperation'; Doring, 'From Bullets to Boreholes'.
36. Doring, 'From Bullets to Boreholes'; Gizelis and Wooden, 'Water Resources, Institutions, & Intrastate Conflict'; Brochmann, 'Signing River Treaties'.
37. Delay and Piou, 'Mutual Aid'; Diekert and Brekke, 'Groups Discipline Resource Use Under Scarcity'.
38. Pathak, 'Ecological Footprints of War'.
39. Pearl, 'Seven Tools of Causal Inference and Machine Learning'.
40. Hegre and Sambanis, 'Sensitivity Analysis of Empirical Results on Civil War Onset'; Miguel et al., 'Economic Shocks and Civil Conflict'; Fearon and Laitin, 'Ethnicity, Insurgency, and Civil War'.
41. Fearon and Laitin, 'Ethnicity, Insurgency, and Civil War'.
42. Collier and Hoeffler, 'Resource Rents, Governance, and Conflict'.
43. Andersson and Lindroth, 'Ecologically Unsustainable Trade'; Muradian and Martinez-Alier, 'Trade and Environment'; Jorgenson and Rice, 'Structural Dynamics of International Trade and Material Consumption'.
44. Bernauer et al., 'Environmental Changes and Violent Conflict'.
45. Davenport, *State Repression and the Domestic Democratic Peace*; Poe and Tate, 'Repression of Human Rights to Personal Integrity in the 1980s'; Rummel, *Power Kills*; Hegre, 'Democracy and Armed Conflict'.
46. Mason, *Caught in the Crossfire*; Fein, 'More Murder in the Middle'; McAdam et al., *Dynamics of Contention*; McCarthy and Zald, 'Resource Mobilization and Social Movements'.

47. Hegre et al., 'Toward a Democratic Civil Peace'; Abulof and Goldman, 'The Domestic Democratic Peace in the Middle East'.
48. Jones and Lupu, 'Is There More Violence in the Middle?'
49. Vesco et al., 'Natural Resources and Conflict'; Lacina, 'Explaining the Severity of Civil Wars'; Lu and Thies, 'Economic Grievance and the Severity of Civil War'.
50. Wimmer et al., 'Ethnic Politics and Armed Conflict'; Cederman et al., 'Why Do Ethnic Groups Rebel?'
51. Cappelli et al., 'Local Vulnerabilities to Climate Change and Conflict in East Africa'; Neudert et al., 'Understanding Causes of Conflict Over Common Village Pastures'.
52. Mendelsohn et al., 'Country-Specific Market Impacts of Climate Change'; Deschênes and Greenstone, 'The Economic Impacts of Climate Change'; Barrios et al., 'Trends in Rainfall and Economic Growth in Africa'.
53. Fearon and Laitin, 'Ethnicity, Insurgency, and Civil War'; Collier and Hoeffler, 'Resource Rents, Governance, and Conflict'.
54. Madsen et al., 'Does Democracy Drive Income in the World, 1500–2000?'; Knutsen, 'Democracy and Economic Growth'; Baum and Lake, 'The Political Economy of Growth'.
55. Hsiang et al., 'Quantifying the Influence of Climate on Human Conflict'.
56. Xie et al., 'The Impacts of Climate Change on Violent Conflict Risk'.
57. Tamburino and Bravo, 'Reconciling Ecology and Human Development'; Global Footprint Network, National Footprint Accounts.
58. Global Footprint Network, National Footprint Accounts.
59. Tamburino and Bravo, 'Reconciling Ecology and Human Development'.
60. Sundberg and Melander, 'Introducing the UCDP Georeferenced Event Dataset'; Davies Shawn, 'Organized Violence since 1989 and the Return of Interstate Conflict'.
61. Coppedge et al., *V-Dem [Country-Year/Country-Date] Dataset v14*.
62. The World Bank, World Bank Open Data.
63. Tamburino and Bravo, 'Reconciling Ecology and Human Development'.
64. Pettersson, *UCDP Battle-Related Deaths Dataset Codebook v 21.1*.
65. Abazi, 'Intrastate Conflict, International Intervention, and Security'.
66. Tamburino and Bravo, 'Reconciling Ecology and Human Development'.
67. Global Footprint Network, 'Glossary'.
68. The World Bank, World Bank Open Data.
69. Global Footprint Network, National Footprint Accounts.
70. Kelley et al., 'Climate Change and the Syrian Drought'; Homer-Dixon et al., 'Synchronous Failure'.
71. Coppedge et al., *V-Dem [Country-Year/Country-Date] Dataset v14*.
72. Luhrmann et al., 'State of the World 2018'.
73. Xie et al., 'The Impacts of Climate Change on Violent Conflict Risk'.
74. Kruschke, *Doing Bayesian Data Analysis*; Burkner, 'brms: A R Package for Bayesian Multilevel Models Using Stan'.
75. R Core Team, *R: A Language and Environment for Statistical Computing*.
76. Burkner, 'brms: A R Package for Bayesian Multilevel Models Using Stan'.
77. Roughly equivalent to a *t*-test in a frequentist framework. Rouder et al., 'Bayesian t Tests for Accepting and Rejecting the Null Hypothesis'.
78. Following Kruschke, *Doing Bayesian Data Analysis*, we consider 'positive evidence' in favour of the tested hypothesis a Bayes Factors (hereafter BF) ≥ 10 and 'strong evidence' a BF ≥ 30.
79. Tamburino and Bravo, 'Reconciling Ecology and Human Development'.
80. Arezki and Bruckner, 'Oil Rents, Corruption, and State Stability'; Gleick, 'Water and Conflict'; Lagi et al., *Food Crises and Political Instability in MENA*.
81. Natalini et al., 'Global Food Security and Food Riots'; Koren et al., 'Food and Water Insecurity as Causes of Social Unrest'.
82. Natalini et al., 'Fuel Riots'.

83. Raleigh et al., 'The Devil is in the Details'; Natalini et al., 'Political Fragility, Food Prices, and Food Riots'; Natalini et al., 'Global Food Security and Food Riots'; Vesco et al., 'Natural Resources and Conflict'.
84. Humphreys, 'Natural Resources, Conflict, and Conflict Resolution'; Collier and Hoeffler, 'Resource Rents, Governance, and Conflict'.
85. Davenport, *State Repression and the Domestic Democratic Peace*; Poe and Tate, 'Repression of Human Rights to Personal Integrity in the 1980s'; Zanger, 'Political Regime Change and Life Integrity Violations'; Cleary, 'Democracy and Indigenous Rebellion in Latin America'.
86. Hegre, 'Toward a Democratic Civil Peace'; Muller and Weede, 'Cross-National Variation in Political Violence'.
87. Jones and Lupu, 'Is There More Violence in the Middle?'
88. Wimmer et al., 'Ethnic Politics and Armed Conflict'; Cederman et al., 'Why Do Ethnic Groups Rebel?'
89. Cappelli et al., 'Local Vulnerabilities to Climate Change and Conflict in East Africa'; Neudert et al., 'Understanding Causes of Conflict Over Common Village Pastures'.
90. Fearon and Laitin, 'Ethnicity, Insurgency, and Civil War'; Collier and Hoeffler, 'Resource Rents, Governance, and Conflict'.
91. Sundberg and Melander, 'Introducing the UCDP Georeferenced Event Dataset'; Davies, 'Organized Violence 1989–2022 and the Return of Conflicts Between States?'
92. Vesco et al., 'Natural Resources and Conflict'.
93. UNDRR, *Crisis Mapping and Disaster Risk Reduction*.
94. The World Bank, World Bank Open Data.
95. Fearon, 'Ethnic and Cultural Diversity by Country'.
96. United Nations Department of Economic and Social Affairs, *Total Number of International Immigrants*.
97. World Inequality Lab (editorial board), *World Inequality Database*.
98. Coppedge et al., *V-Dem [Country-Year/Country-Date] Dataset v14*.
99. Pearl, 'Seven Tools of Causal Inference and Machine Learning'.
100. Scutari, 'Learning Bayesian Networks with the bnlearn R Package'.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors

Elizaveta Kopacheva holds a PhD in Political Science from Linnaeus University (Sweden), and her work lies in in the field of computational social science. Her research has examined political participation, with particular emphasis on non-institutionalised forms such as protest and online activism, as well as political mobilisation and social inequalities in civic engagement. Her work integrates political science theory with advanced quantitative and computational methods, including statistical modelling, Bayesian analysis, machine learning, text mining, and social network analysis. She has been involved in numerous interdisciplinary projects at the intersection of social science and data science, addressing topics such as opinion polarisation, electoral behaviour, digital political communication, and democratic resilience.

Giangiacomo Bravo is professor in social sciences at Linnaeus University, Sweden. His research focuses on the relation between human behavior and the environment and on the management of common resources.

Dr Davide Natalini is Senior Research Fellow at the Global Sustainability Institute based at Anglia Ruskin University, in the UK. Davide is an interdisciplinary environmental social scientist who studies complex social-ecological systems. His research focusses on how humans and nature interact, with specific attention to environmental conflict, systemic risks and community participation. In his research he implements a range of mixed methods such as computer simulation, quantitative, qualitative and participatory methods.

ORCID

Elizaveta Kopacheva http://orcid.org/0000-0002-3708-6333
Giangiacomo Bravo http://orcid.org/0000-0003-2837-0137
Davide Natalini http://orcid.org/0000-0001-6975-5522

Use of generative artificial intelligence

The Consensus.app was used to help searching and classifying relevant literature. Writefull.com was used for language improvement.

References

Abazi, Enika, 2001. 'Intrastate Conflicts, International Interventions and Their Implications on Security Issues: Case of Kosovo'. *COPRI Working Papers, Copenhagen Peace Research Institute* 32(ffhalshs–01340929f), 1–34.

Abel, Guy J., Michael Brottrager, Jesus Crespo Cuaresma and Raya Muttarak, 2019. 'Climate, Conflict and Forced Migration'. *Global Environmental Change* 54, 239–249.

Abulof, Uriel and Ogen Goldman, 2015. 'The Domestic Democratic Peace in the Middle East'. *International Journal of Conflict & Violence* 9(1).

Acemoglu, Daron, Mikhail Golosov, Aleh Tsyvinski and Pierre Yared, 2012. 'A Dynamic Theory of Resource Wars'. *The Quarterly Journal of Economics* 127(1), 283–331.

Adams, Courtland, Tobias Ide, Jon Barnett and Adrien Detges, 2018. 'Sampling Bias in Climate–Conflict Research'. *Nature Climate Change* 8(3), 200–203.

Adanu, Kwami, 2023. 'Population, Institutions, and Violent Conflicts – How Important is Population Pressure in Violent Resource-Based Conflicts?' *Peace Economics, Peace Science and Public Policy* 29(3), 249–277.

Andersson, Jan Otto and Mattias Lindroth, 2001. 'Ecologically Unsustainable Trade'. *Ecological Economics* 37(1), 113–122.

Arbatlı, Cemal Eren, Quamrul H. Ashraf, Oded Galor and Marc Klemp, 2015. 'Diversity and Conflict'. *Political Economy — Development: Underdevelopment & Poverty eJournal.*

Arezki, Rabah and Markus Bruckner, 2011. 'Oil Rents, Corruption, and State Stability: Evidence from Panel Data Regressions'. *European Economic Review* 55(7), 955–963.

Baalen, Sebastian van and Malin Mobjork, 2018. 'Climate Change and Violent Conflict in East Africa: Integrating Qualitative and Quantitative Research to Probe the Mechanisms'. *International Studies Review* 20(4), 547–575.

Barrios, Salvador, Luisito Bertinelli and Eric Strobl, 2010. 'Trends in Rainfall and Economic Growth in Africa: A Neglected Cause of the African Growth Tragedy'. *The Review of Economics and Statistics* 92(2), 350–366.

Baum, Matthew A. and David A. Lake, 2003. 'The Political Economy of Growth: Democracy and Human Capital'. *American Journal of Political Science* 47(2), 333–347.

Bayramov, Agha, 2018. 'Review: Dubious Nexus Between Natural Resources and Conflict'. *Journal of Eurasian Studies* 9(1), 72–81.

Bernauer, T., Tobias Bohmelt and V. Koubi, 2012. 'Environmental Changes and Violent Conflict'. *Environmental Research Letters* 7(1), 015601.

Brancati, Dawn, 2007. 'Political Aftershocks: The Impact of Earthquakes on Intrastate Conflict'. *Journal of Conflict Resolution* 51(5), 715–743.

Brochmann, Marit, 2012. 'Signing River Treaties — Does it Improve River Cooperation?' *International Interactions* 38(2), 141–163.
Burkner, Paul-Christian, 2017. 'Brms: An R Package for Bayesian Multilevel Models Using Stan'. *Journal of Statistical Software* 80(1).
Cappelli, Federica, Valeria Costantini, Mariagrazia D'Angeli, Giovanni Marin and Elena Paglialunga, 2024. 'Local Sources of Vulnerability to Climate Change and Armed Conflicts in East Africa'. *Journal of Environmental Management* 355, 120403.
Cederman, Lars-Erik, Andreas Wimmer and Brian Min, 2010. 'Why Do Ethnic Groups Rebel? New Data and Analysis' [in eng]. *World Politics (Cambridge, UK)* 62(1), 87–119.
Chisadza, Carolyn, Matthew W. Clance, Rangan Gupta and M. Wohar, 2019. 'Giant Oil Discoveries and Conflicts'. *Environment, Development and Sustainability*, 1–30.
Cleary, Matthew R., 2000. 'Democracy and Indigenous Rebellion in Latin America'. *Comparative Political Studies* 33(9), 1123–1153.
Collier, Paul and Anke Hoeffler, 2005. 'Resource Rents, Governance, and Conflict'. *Journal of Conflict Resolution* 49(4), 625–633.
Conrad, Justin M., Kevin T. Greene, James Igoe Walsh and Beth Elise Whitaker, 2019. 'Rebel Natural Resource Exploitation and Conflict Duration'. *Journal of Conflict Resolution* 63(3), 591–616.
Coppedge, Michael, John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, Jan Teorell, Kyle L. Marquardt, Juraj Medzihorsky, Daniel Pemstein, Linnea Fox, Lisa Gastaldi, Josefine Pernes, Oskar Rydén, Johannes von Römer, Eitan Tzelgov, Yi-ting Wang, and Steven Lloyd Wilson, 2024 '*V-Dem [Country-Year/country-Date] Dataset v14*'. Varieties of Democracy (V-Dem) Project.
Davenport, Christian, 2007. *State Repression and the Domestic Democratic Peace*. Cambridge University Press, Cambridge.
Davies, Shawn, Therese Pettersson and Magnus Oberg, 2023. 'Organized Violence 1989–2022¨ and the Return of Conflicts Between States?' *Journal of Peace Research* 60(4), 809–825.
Delay, E. and C. Piou, 2019. 'Mutual Aid: When Does Resource Scarcity Favour Group Cooperation?' *Ecological Complexity* 40, 100790.
Denly, Michael, Michael G. Findley, J. Hall, Andrew Stravers and J. I. Walsh, 2022. 'Do Natural Resources Really Cause Civil Conflict? Evidence from the New Global Resources Dataset'. *Journal of Conflict Resolution* 66(3), 387–412.
Deschênes, Olivier and Michael Greenstone, 2007. 'The Economic Impacts of Climate Change: Evidence from Agricultural Output and Random Fluctuations in Weather'. *American Economic Review* 97(1), 354–385.
Diekert, F. and K. Brekke, 2021. 'Groups Discipline Resource Use Under Scarcity'. *Theory and Decision* 92(1), 75–103.
Doring, S., 2020. 'From Bullets to Boreholes: A Disaggregated Analysis of Domestic Water¨ Cooperation in Drought-Prone Regions'. *Global Environmental Change* 65, 102147.
Durham, William H., 1979. *Scarcity and Survival in Central America: Ecological Origins of the Soccer War*. Stanford University Press, Stanford.
Fearon, James D., 2003. 'Ethnic and Cultural Diversity by Country'. *Journal of Economic Growth* 8 (2), 195–222.
Fearon, James D. and David D. Laitin, 2003. 'Ethnicity, Insurgency, and Civil War'. *The American Political Science Review* 97(1), 75–90.
Fein, Helen., 1995. 'More Murder in the Middle: Life-Integrity Violations and Democracy in the World, 1987'. *Human Rights Quarterly* 17(1), 170–191.
Gargano, Francesco, Lucia Tamburino, Fabio Bagarello and Giangiacomo Bravo, 2017. 'Largescale Effects of Migration and Conflict in Pre-Agricultural Groups: Insights from a Dynamic Model' [Edited by John P. Hart]. *PLOS ONE* 12(3), e0172262.
Gawande, Kishore, Devesh Kapur and Shanker Satyanath, 2017. 'Renewable Natural Resource Shocks and Conflict Intensity: Findings from India's Ongoing Maoist Insurgency'. *Journal of Conflict Resolution* 61(1), 140–172.

Gizelis, Theodora-Ismene and Amanda E. Wooden, 2010. 'Water Resources, Institutions, & Intrastate Conflict'. *Political Geography* 29(8), 444–453.

Gleick, Peter H., 1993. 'Water and Conflict: Fresh Water Resources and International Security'. *International Security* 18(1), 79–112.

Global Footprint Network, 'Glossary'. 2020. https://www.footprintnetwork.org/resources/glossary/ [Accessed 3 January 2020].

Global Footprint Network, 2024. National Footprint Accounts. 2024 ed. Global Footprint Network, Oakland, CA. http://data.footprintnetwork.org

Hawes, Gary, 1990. 'Theories of Peasant Revolution: A Critique and Contribution from the Philippines'. *World Politics* 42(2), 261–298.

Hegre, Havard, 2001. 'Toward a Democratic Civil Peace? Democracy, Political Change, and Civil War, 1816–1992'. *American Political Science Review* 95(1), 33–48.

Hegre, Havard, 2014. 'Democracy and Armed conflict'. *Journal of Peace Research* 51(2), 159–172.

Hegre, Havard, Tanja Ellingsen, Scott Gates and Nils Petter Gleditsch, 2001. 'Toward a° Democratic Civil Peace? Democracy, Political Change, and Civil War, 1816-1992'. *The American Political Science Review* 95(1), 33–48. http://www.jstor.org/stable/3117627

Hegre, Havard and Nicholas Sambanis, 2006. 'Sensitivity Analysis of Empirical Results° on Civil War Onset'. *Journal of Conflict Resolution* 50(4), 508–535.

Homer-Dixon, Thomas F., 1999. *Environment, Scarcity, and Violence*. Princeton University Press, Princeton.

Homer-Dixon, Thomas, Brian Walker, Reinette Biggs, Anne-Sophie Crepin, Carl Folke, Eric F. Lambin Garry D. Peterson, J. Rockström, M. Scheffer, W. Steffen and M. Troell, 2015. 'Synchronous Failure: The Emerging Causal Architecture of Global Crisis'. *Ecology and Society* 20(3).

Hsiang, Solomon M., Marshall Burke and Edward Miguel, 2013. 'Quantifying the Influence of Climate on Human Conflict'. *Science* 341(6151).

Humphreys, Macartan, 2005. 'Natural Resources, Conflict, and Conflict Resolution: Uncovering the Mechanisms'. *Journal of Conflict Resolution* 49(4), 508–537.

Ide, Tobias, 2023. 'Rise or Recede? How Climate Disasters Affect Armed Conflict Intensity'. *International Security* 47(4), 50–78.

Jones, Zachary M. and Yonatan Lupu, 2018. 'Is There More Violence in the Middle? [In Eng]'. *American Journal of Political Science (HOBOKEN)* 62(3), 652–667.

Jorgenson, Andrew K. and James Rice, 2005. 'Structural Dynamics of International Trade and Material Consumption: A Cross-National Study of the Ecological Footprints of Less-Developed Countries'. *Journal of World-Systems Research* 11(1), 57–77.

Kahl, Colin H., 2018. *States, Scarcity, and Civil Strife in the Developing World*. Princeton University Press, Princeton, NJ.

Kelley, Colin P., Shahrzad Mohtadi, Mark A. Cane, Richard Seager and Yochanan Kushnir, 2015. 'Climate Change in the Fertile Crescent and Implications of the Recent Syrian Drought'. *Proceedings of the National Academy of Sciences* 112(11), 3241–3246.

Knutsen, Carl Henrik, 2012. 'Democracy and Economic Growth: A Survey of Arguments and Results'. *International Area Studies Review* 15(4), 393–415.

Koren, Ore, Benjamin E Bagozzi and Thomas S Benson, 2021. 'Food and Water Insecurity as Causes of Social Unrest: Evidence from Geolocated Twitter Data'. *Journal of Peace Research* 58 (1), 67–82.

Koubi, Vally and Gabriele Spilker, 2017. 'Natural Resources, Climate Change, and Conflict'. June. https://oxfordre.com/politics/view/10.1093/acrefore/9780190228637.001.0001/acrefore-9780190228637-e-346

Koubi, Vally, Gabriele Spilker, Tobias Bohmelt and Thomas Bernauer, 2014. 'Do natural¨ Resources Matter for Interstate and Intrastate Armed Conflict?' *Journal of Peace Research* 51 (2), 227–243.

Kruschke, John, 2014. *Doing Bayesian Data Analysis: A Tutorial with R, JAGS, and Stan*. Academic Press, Amsterdam.

Lacina, Bethany, 2006. 'Explaining the Severity of Civil Wars'. *Journal of Conflict Resolution* 50(2), 276–289.

Lagi, Marco, Karla Z. Bertrand and Yaneer Bar-Yam, 2011. 'The Food Crises and Political Instability in North Africa and the Middle East'. http://doi.org/10.48550/ARXIV.1108.2455

Levy, Marc A., 1995. 'Is the Environment a National Security Issue?' *International Security* 20(2), 35–62.

Lu, Lingyu and Cameron G Thies, 2011. 'Economic Grievance and the Severity of Civil War'. *Civil Wars* 13(3), 215–231.

Luhrmann, Anna, Sandra Grahn, Richard Morgan, Shreeya Pillai and Staffan I. Lind-¨ Berg, 2019. 'State of the World 2018: Democracy Facing Global Challenges'. *Democratization* 26(6), 895–915.

Madsen, Jakob B., Paul A. Raschky and Ahmed Skali, 2015. 'Does Democracy Drive Income in the World, 1500–2000?' *European Economic Review* 78, 175–195.

Mason, David T., 2004. *Caught in the Crossfire: Revolution, Repression, and the Rational Peasant*. Rowman & Littlefield Publishers, Lanham.

McAdam, Doug, Sidney Tarrow and Charles Tilly, 2001. *Dynamics of Contention*. Cambridge University Press, New York.

McCarthy, John D. and Mayer N. Zald, 1977. 'Resource Mobilization and Social Movements: A Partial Theory'. *American Journal of Sociology* 82(6), 1212–1241.

McLaughlin Mitchell, Sara, 2024. 'Cross-Border Troubles? Interstate River Conflicts and Intrastate Violence'. *Political Geography* 111, 103109.

Mendelsohn, Robert, Wendy Morrison, Michael Schlesinger and Natalia Andronova, 2000. 'Country-Specific Market Impacts of Climate Change'. *Climatic Change* 45(3–4), 553–569.

Miguel, Edward, Shanker Satyanath and Ernest Sergenti, 2004. 'Economic Shocks and Civil Conflict: An Instrumental Variables Approach'. *Journal of Political Economy* 112(4), 725–753.

Mildner, Stormy-Annika, Gitta Lauster and Wiebke Wodni, 2011. 'Scarcity and Abundance Revisited: A Literature Review on Natural Resources and Conflict'. *International Journal of Conflict and Violence (IJCV)* 5(1), 155–172.

Muller, Edward N. and Erich Weede, 1990. 'Cross-National Variation in Political Violence: A Rational Action Approach'. *Journal of Conflict Resolution* 34(4), 624–651.

Muradian, R. and J. Martinez-Alier, 2001. 'Trade and Environment: From a 'Southern' Perspective'. *Ecological Economics* 36(2), 281–297.

Natalini, Davide, Giangiacomo Bravo and Aled Wynne Jones, 2019. 'Global Food Security and Food Riots – an Agent-Based Modelling Approach'. *Food Security* 11(5), 1153–1173.

Natalini, Davide, Giangiacomo Bravo and Edward Newman, 2020. 'Fuel Riots: Definition, Evidence and Policy Implications for a New Type of Energy-Related Conflict'. *Energy Policy* 147, 111885.

Natalini, Davide, Aled Wynne Jones and Giangiacomo Bravo, 2015. 'Quantitative Assessment of Political Fragility Indices and Food Prices as Indicators of Food Riots in Countries'. *Sustainability* 7(4), 4360–4385.

Neudert, Regina, Insa Theesfeld, Alexandre Didebulidze, Naiba Allahverdiyeva and Volker Beckmann, 2020. 'Understanding Causes of Conflict Over Common Village Pastures – a Comparative Analysis of Property Rights in Azerbaijan and Georgia'. *Society & Natural Resources* 33(3), 347–367.

Nie, Martin, 2003. 'Drivers of Natural Resource-Based Political Conflict'. *Policy Sciences* 36(3–4), 307–341.

Nie, Z., Xiaojun Yang and Q. Tu, 2020. 'Resource Scarcity and Cooperation: Evidence from a Gravity Irrigation System in China'. *World Development* 135, 105035.

O'Brochta, William, 2019. 'A Meta-Analysis of Natural Resources and Conflict'. *Research & Politics* 6(1), 2053168018818232.

Oneal, John R and Bruce M Russet, 1997. 'The Classical Liberals Were Right: Democracy, Interdependence, and Conflict, 1950–1985'. *International Studies Quarterly* 41(2), 267–293.

Pathak, Swapna, 2020. 'Ecological Footprints of War: An Exploratory Assessment of the Longterm Impact of Violent Conflicts on National Biocapacity from 1962–2009'. *Journal of Environmental Studies and Sciences* 10(4), 380–393.

Pearl, Judea, 2019. 'The Seven Tools of Causal Inference, with Reflections on Machine Learning'. *Communications of the ACM* 62(3), 54–60.

Pettersson, Therese, 2021. 'UCDP Battle-Related Deaths Dataset Codebook V 21.1'. https://ucdp.uu.se/downloads/

Poe, Steven C. and C. Neal Tate, 1994. 'Repression of Human Rights to Personal Integrity in the 1980s: A Global Analysis'. *American Political Science Review* 88(4), 853–872. https://EconPapers.repec.org/RePEc:cup:apsrev:v:88:y:1994:i:04:p:853-87209

Raleigh, Clionadh, Hyun Jin Choi and Dominic Kniveton, 2015. 'The Devil is in the Details: An Investigation of the Relationships Between Conflict, Food Price and Climate Across Africa'. *Global Environmental Change* 32, 187–199.

R Core Team, 2024. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/

Reuveny, Rafael, 2007. 'Climate Change-Induced Migration and Violent Conflict'. *Political Geography* 26(6), 656–673.

Rigterink, Anouk S., 2020. 'Diamonds, Rebel's and Farmer's Best Friend: Impact of Variation in the Price of a Lootable, Labor-Intensive Natural Resource on the Intensity of Violent Conflict'. *Journal of Conflict Resolution* 64(1), 90–126.

Rouder, Jeffrey N., Paul L. Speckman, Dongchu Sun, Richard D. Morey and Geoffrey Iverson, 2009. 'Bayesian T Tests for Accepting and Rejecting the Null Hypothesis'. *Psychonomic Bulletin & Review* 16(2), 225–237.

Rummel, Rudolph J., 1997. *Power Kills: Democracy as a Method of Nonviolence*. Transaction Books, New Brunswick.

Schleussner, Carl-Friedrich, Jonathan F. Donges, Reik V. Donner and Hans Joachim Schellnhuber, 2016. 'Armed-Conflict Risks Enhanced by Climate-Related Disasters in Ethnically Fractionalized Countries'. *Proceedings of the National Academy of Sciences* 113(33), 9216–9221.

Schleussner, C., J. Donges, R. Donner and H. Schellnhuber, 2016. 'Armed-Conflict Risks Enhanced by Climate-Related Disasters in Ethnically Fractionalized Countries'. *Proceedings of the National Academy of Sciences* 113(33), 9216–9221.

Scutari, Marco, 2010. 'Learning Bayesian Networks with the Bnlearn R Package'. *Journal of Statistical Software* 35(3), 1–22. http://www.jstatsoft.org/v35/i03/

Snow, Donald M., 1996. *Uncivil Wars: International Security and the New Internal Conflicts*. Lynne Rienner Publishers, Boulder, CO, US.

Sorens, Jason, 2011. 'Mineral Production, Territory, and Ethnic Rebellion: The Role of Rebel Constituencies'. *Journal of Peace Research* 48(5), 571–585.

Steffen, Will, Katherine Richardson, Johan Rockstrom, Sarah E. Cornell, Ingo Fetzer, Elena M. Bennett Reinette Biggs, S. R. Carpenter, W. De Vries, C. A. De Wit and C. Folke, 2015. 'Planetary Boundaries: Guiding Human Development on a Changing Planet'. *Science* 347(6223), 736.

Sundberg, Ralph and Erik Melander, 2013. 'Introducing the UCDP Georeferenced Event Dataset'. *Journal of Peace Research* 50(4), 523–532.

Sweijs, Tim, Marleen De Haan and Hugo Van Manen, 2022. *Unpacking the Climate Security Nexus: Seven Pathologies Linking Climate Change to Violent Conflict*. Hague Centre for Strategic Studies.

Tamburino, Lucia and Giangiacomo Bravo, 2021. 'Reconciling a Positive Ecological Balance with Human Development: A Quantitative Assessment'. *Ecological Indicators* 129, 107973.

Transaction Books, 1997. https://books.google.se/books?id=WyEuDwAAQBAJ

UNDRR, 2021. 'Crisis Mapping and Disaster Risk Reduction'. https://www.undrr.org/news/crisis-

mapping-and-disaster-risk-reduction
United Nations Department of Economic and Social Affairs, 2025. 'Total Number of International Immigrants'. Data provided by the United Nations Department of Economic and Social Affairs, with minor processing by Our World in Data. [Accessed 9 January 2024]. https://ourworldindata.org/migration
Vesco, Paola, Shouro Dasgupta, Enrica De Cian and Carlo Carraro, 2020. 'Natural Resources and Conflict: A Meta-Analysis of the Empirical Literature'. *Ecological Economics* 172, 106633.
Wackernagel, Mathis, Alessandro Galli, Laurel Hanscom, David Lin, Laetitia Mailhes and Tony Drummond, 2018. 'Ecological Footprint Accounts: Criticisms and Applications'. In *Routledge Handbook of Sustainability Indicators and Indices*, eds. S. Bell and S. Morse. Routledge, London, 521–539.
Wallensteen, Peter., 1984. 'Universalism Vs. Particularism: On the Limits of Major Power Order'. *Journal of Peace Research* 21(3), 243–257.
Wimmer, Andreas, Lars-Erik Cederman and Brian Min, 2009. 'Ethnic Politics and Armed Conflict: A Configurational Analysis of a New Global Data Set' [in eng]. *American Sociological Review (Los Angeles, CA)* 74(2), 316–337.
Wischnath, Gerdis and Halvard Buhaug, 2014. 'Rice or Riots: On Food Production and Conflict Severity Across India'. *Political Geography* 43, 6–15.
The World Bank, 2024. 'World Bank Open Data'. https://data.worldbank.org/
World Inequality Lab (editorial board), 2025. 'World Inequality Database'. [Accessed 10 July 2025]. https://wid.world/data/
Xie, Xiaolan, Mengmeng Hao, Fangyu Ding, J. Scheffran, Tobias Ide, Jean-Franc¸ois Maystadt Yushu Qian, Q. Wang, S. Chen, J. Wu and K. Sun, 2024. 'The Impacts of Climate Change on Violent Conflict Risk: A Review of Causal Pathways'. *Environmental Research Communications* 6(11), 112002.
Zanger, Sabine C., 2000. 'A Global Analysis of the Effect of Political Regime Changes on Life Integrity Violations, 1977-93'. *Journal of Peace Research* 37(2), 213–233.

Appendix: A Robustness test results

Log-transformed conflict indicators

This robustness test examines the effects of predictors on conflict outcomes, with all continuous predictors – specifically, PB and income – being log-transformed to reduce skewness.

The model structure is as follows:

$$\text{Income}_{i,j} = \beta_0 + \beta_1 \times \text{PB}_{i,j} + \beta_2 \times \text{Political regime}_{i,j} + u_j^{(1)} + \varepsilon_{i,j}^{(1)};$$

$$\text{Conflict}_{i,j} = \alpha_0 + \alpha_1 \times \text{PB}_{i,j} + \alpha_2 \times \text{Political regime}_{i,j} + \alpha_3 \times \text{Income}_{i,j} + u_j^{(2)} + \varepsilon_{i,j}^{(2)},$$

where $\text{Income}_{i,j}$, $\text{PB}_{i,j}$ and $\text{Political regime}_{i,j}$ are the income level, PB and political regime of country j in year i, respectively; $\text{Conflict}_{i,j}$ is the presence of conflict in country j in year i; $u_j^{(1)}$ and $u_j^{(2)}$ are the random effects of each country; and $\varepsilon_{ij}^{(1)}$ and $\varepsilon_{ij}^{(2)}$ are the error terms.

The same model specification is applied when using conflict intensity (measured by casualties) as the main dependent variable.

The presence of conflict models were estimated using a Bernoulli likelihood, while the conflict intensity (casualties) models – Student-t likelihood.

Table A1. Coefficient estimates for the conflict model.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.770	[8.574; 8.966]	40000:1
Income	PB	0.230	[0.172; 0.288]	40000:1
Income	Regime: EA	0.275	[0.231; 0.318]	40000:1
Income	Regime: ED	0.360	[0.31; 0.409]	40000:1
Income	Regime: LD	0.449	[0.373; 0.525]	40000:1
Conflict	(Intercept)	−3.569	[−4.599; −2.635]	1:40000
Conflict	PB	1.102	[0.465; 1.769]	4999:1
Conflict	Income	−1.130	[−1.636; −0.631]	1:40000
Conflict	Regime: EA	−0.494	[−0.957; −0.032]	1:54
Conflict	Regime: ED	−1.297	[−1.887; −0.716]	1:40000
Conflict	Regime: LD	−3.687	[−6.388; −1.399]	1:1537

Note: The model includes 4343 observations from 163 countries. Income (GDP per capita, PPP) and PB were log-transformed for normalization and then standardized for estimation.

Table A2. Coefficient estimates for the casualties model.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	7.928	[7.616; 8.242]	40000:1
Income	PB	0.156	[0.077; 0.233]	19999:1
Income	Regime: EA	0.343	[0.272; 0.415]	40000:1
Income	Regime: ED	0.200	[0.095; 0.307]	13332:1
Income	Regime: LD	2.612	[1.177; 4.052]	3332:1
Casualties	(Intercept)	5.926	[5.472; 6.372]	40000:1
Casualties	PB	0.474	[0.209; 0.745]	3635:1
Casualties	Income	−0.202	[−0.504; 0.096]	1:10
Casualties	Regime: EA	−0.290	[−0.636; 0.056]	1:19
Casualties	Regime: ED	−0.273	[−0.749; 0.2]	1:7
Casualties	Regime: LD	−0.632	[−2.361; 1.152]	1:3

Note: The model includes 667 observations from 44 countries. Income (GDP per capita, PPP) was logtransformed for normalisation and then standardized for estimation; PB was standardized for estimation; casualties was log-transformed for normalisation.

Delayed Effects of Predictors on Conflict

This robustness test examines whether PB measured in previous years (t −1 and t −2) affects conflict outcomes observed at time t. The model structure is as follows:

$$\text{Income}_{i,j} = \beta_0 + \beta_1 \times \text{PB}_{i-n,j} + \beta_2 \times \text{Political regime}_{i,j} + u_j^{(1)} + \varepsilon_{i,j}^{(1)};$$

$$\text{Conflict}_{i,j} = \alpha_0 + \alpha_1 \times \text{PB}_{i-n,j} + \alpha_2 \times \text{Political regime}_{i,j} + \alpha_3 \times \text{Income}_{i,j} + u_j^{(2)} + \varepsilon_{i,j}^{(2)},$$

where Income$_{i,j}$, Political regime$_{i,j}$ and Conflict$_{i,j}$ are the income level, political regime and the presence of conflict in country j in year i, respectively; PB$_{i-n,j}$ is the PB in country j in year i-n with $n = 1$ (when studying the effect of PB on conflict one year later) and 2 (for conflict 2 years later); $u_j^{(1)}$ and $u_j^{(2)}$ are the random effects of each country; and $\varepsilon_{ij}^{(1)}$ and $\varepsilon_{ij}^{(2)}$ are the error terms.

The same model specification is applied when using conflict intensity (measured by casualties) as the main dependent variable.

The presence of conflict models were estimated using a Bernoulli likelihood, while the conflict intensity (casualties) models – Student-t likelihood.

The following four tables report the estimation results for this robustness test.

Table A3. Coefficient estimates for the conflict model.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.781	[8.587; 8.971]	40000:1
Conflict	(Intercept)	−3.595	[−4.618; −2.653]	1:40000
Income	PB (previous year)	0.219	[0.159; 0.278]	40000:1
Income	Regime: EA	0.274	[0.229; 0.318]	40000:1
Income	Regime: ED	0.356	[0.306; 0.406]	40000:1
Income	Regime: LD	0.426	[0.351; 0.503]	40000:1
Conflict	(Intercept)	−3.595	[−4.618; −2.653]	1:40000
Conflict	PB (previous year)	1.008	[0.379; 1.672]	1249:1
Conflict	Income	−1.173	[−1.696; −0.657]	1:40000
Conflict	Regime: EA	−0.431	[−0.913; 0.047]	1:25
Conflict	Regime: ED	−1.220	[−1.836; −0.619]	1:39999
Conflict	Regime: LD	−3.609	[−6.301; −1.314]	1:1332

Note: The model includes 4197 observations from 163 countries. Income (GDP per capita, PPP) and PB were logtransformed for normalization and then standardized for estimation. PB is measured at time t −1, while all the other variables at time t.

Table A4. Coefficient estimates for the casualties model.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	7.949	[7.625; 8.27]	40000:1
Income	PB (previous year)	0.171	[0.094; 0.249]	40000:1
Income	Regime: EA	0.337	[0.264; 0.411]	40000:1
Income	Regime: ED	0.180	[0.071; 0.289]	1480:1
Income	Regime: LD	2.579	[1.125; 4.044]	1599:1
Casualties	(Intercept)	5.946	[5.481; 6.411]	40000:1
Casualties	PB (previous year)	0.266	[−0.008; 0.538]	34:1
Casualties	Income	−0.169	[−0.468; 0.125]	1:7
Casualties	Regime: EA	−0.289	[−0.645; 0.069]	1:17
Casualties	Regime: ED	−0.300	[−0.787; 0.188]	1:8
Casualties	Regime: LD	−0.595	[−2.297; 1.142]	1:3

Note: The model includes 640 observations from 43 countries. Income (GDP per capita, PPP) was logtransformed for normalization and then standardized for estimation. PB was standardized for estimation. PB is measured at time t −1, while all the other variables at time t.

Table A5. Coefficient estimates for the conflict model.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.803	[8.612; 8.997]	40000:1
Income	PB (2 years ago)	0.197	[0.137; 0.256]	40000:1
Income	Regime: EA	0.263	[0.217; 0.309]	40000:1
Income	Regime: ED	0.348	[0.297; 0.399]	40000:1
Income	Regime: LD	0.405	[0.329; 0.482]	40000:1
Conflict	(Intercept)	−3.599	[−4.622; −2.644]	1:40000
Conflict	PB (2 years ago)	0.896	[0.26; 1.551]	344:1
Conflict	Income	−1.189	[−1.721; −0.659]	1:40000
Conflict	Regime: EA	−0.344	[−0.841; 0.149]	1:10
Conflict	Regime: ED	−1.149	[−1.768; −0.531]	1:9999
Conflict	Regime: LD	−3.487	[−6.125; −1.248]	1:850

Note: The model includes 4052 observations from 163 countries. Income (GDP per capita, PPP) and PB were logtransformed for normalization and then standardized for estimation. PB is measured at time *t* −2, while all the other variables at time *t*.

Table A6. Coefficient estimates for the casualties model.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	7.966	[7.637; 8.292]	40000:1
Income	PB (2 years ago)	0.192	[0.108; 0.276]	39999:1
Income	Regime: EA	0.328	[0.254; 0.402]	40000:1
Income	Regime: ED	0.173	[0.065; 0.281]	975:1
Income	Regime: LD	2.542	[1.076; 4.022]	1537:1
Casualties	(Intercept)	5.919	[5.437; 6.396]	40000:1
Casualties	PB (2 years ago)	0.186	[−0.108; 0.47]	8:1
Casualties	Income	−0.199	[−0.522; 0.111]	1:9
Casualties	Regime: EA	−0.274	[−0.646; 0.103]	1:12
Casualties	Regime: ED	−0.288	[−0.787; 0.213]	1:7
Casualties	Regime: LD	−0.470	[−2.219; 1.376]	1:2

Note: The model includes 617 observations from 43 countries. Income (GDP per capita, PPP) was logtransformed for normalisation and then standardised for estimation. PB was standardised for estimation. PB is measured at time *t* −2, while all the other variables at time *t*.

Influence of Burden on the Local Ecosystem on conflict in Countries with Varying Natural Resource Rents

This robustness test examines the effects of predictors on conflict outcomes across countries with different levels of dependence on natural resources. Specifically, we analysed two subsets of countries based on the share of total natural resource rents (as a percentage of GDP):

- Low-rent countries (Q1–) – countries in the lowest quartile of natural resource rents; and
- High-rent countries (Q3+) – countries in the highest quartile of natural resource rents.

Data on natural resource rents were obtained from the World Bank.[94]

The model specification follows the baseline model described in the main text. Presence of conflict models use a Bernoulli likelihood, while conflict-intensity models use a Student-t likelihood. Coefficient estimates for the casualties model in countries with small (Q1-) percentage of total natural resource rents (calculated as percentage of GDP) are not reported due to insufficiency of data (*number of observations*: 101; *number of countries*: 6).

The results of this robustness test are reported in the following tables.

Table A7. Coefficient estimates for the conflict model in countries with small (Q1-) percentage of total natural resource rents (calculated as percentage of GDP).

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	9.553	[9.182; 9.931]	40000:1
Income	PB	0.098	[−0.012; 0.209]	23:1
Income	Regime: EA	0.220	[−0.04; 0.48]	20:1
Income	Regime: ED	0.218	[−0.048; 0.485]	18:1
Income	Regime: LD	0.173	[−0.104; 0.45]	8:1
Conflict	(Intercept)	−13.309	[−24.056; −5.706]	1:40000
Conflict	PB	1.558	[−0.491; 3.937]	13:1
Conflict	Income	−2.188	[−3.7; −0.869]	1:2856
Conflict	Regime: EA	5.119	[−1.609; 14.654]	11:1
Conflict	Regime: ED	3.735	[−3.058; 13.206]	5:1
Conflict	Regime: LD	4.128	[−4.321; 14.38]	4:1

Note: The model includes 1192 observations from 57 countries. Income (GDP per capita, PPP) and PB were log-transformed for normalization and then standardized for estimation.

Table A8. Coefficient estimates for the casualties model in countries with small percentage of total natural resource rents (calculated as percentage of GDP).

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.560	[6.967; 10.156]	13332:1
Income	PB	0.257	[0.116; 0.398]	2352:1
Income	Regime: EA	0.298	[−0.061; 0.659]	18:1
Income	Regime: ED	−0.044	[−0.42; 0.334]	1:1
Income	Regime: LD	2.092	[−0.769; 4.904]	16:1
Casualties	(Intercept)	4.875	[1.726; 8.075]	356:1
Casualties	PB	0.102	[−0.664; 0.899]	2:1
Casualties	Income	−0.265	[−1.333; 1.051]	1:2
Casualties	Regime: EA	2.397	[−0.28; 5.081]	26:1
Casualties	Regime: ED	2.041	[−0.736; 4.8]	14:1
Casualties	Regime: LD	0.490	[−4.158; 4.733]	2:1

Note: The model includes 101 observations from 6 countries. Income (GDP per capita, PPP) was log-transformed for normalization and then standardized for estimation. PB was standardized for estimation.

Table A9. Coefficient estimates for the conflict model in countries with large (Q3+) percentage of total natural resource rents (calculated as percentage of GDP).

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.335	[7.987; 8.679]	40000:1
Income	PB	0.347	[0.25; 0.446]	40000:1
Income	Regime: EA	0.229	[0.162; 0.297]	40000:1
Income	Regime: ED	0.329	[0.231; 0.428]	40000:1
Income	Regime: LD	0.672	[0.494; 0.854]	40000:1
Conflict	(Intercept)	−2.319	[−3.47; −1.286]	1:39999
Conflict	PB	0.980	[0.147; 1.876]	97:1
Conflict	Income	−1.186	[−1.925; −0.45]	1:1025
Conflict	Regime: EA	−0.514	[−1.105; 0.066]	1:23
Conflict	Regime: ED	−0.768	[−1.693; 0.147]	1:19
Conflict	Regime: LD	−80.502	[−351.094; −2.499]	1:178

Note: The model includes 1249 observations from 63 countries. Income (GDP per capita, PPP) and PB were log-transformed for normalization and then standardized for estimation.

Table A10. Coefficient estimates for the casualties model in countries with large percentage of total natural resource rents (calculated as percentage of GDP).

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	7.858	[7.396; 8.322]	40000:1
Income	PB	0.142	[0.049; 0.237]	799:1
Income	Regime: EA	0.195	[0.098; 0.292]	40000:1
Income	Regime: ED	0.325	[0.082; 0.568]	240:1
Casualties	(Intercept)	5.952	[5.341; 6.545]	40000:1
Casualties	PB	0.432	[0.067; 0.808]	100:1
Casualties	Income	−0.440	[−0.907; −0.014]	1:46
Casualties	Regime: EA	−0.863	[−1.33; −0.4]	1:7999
Casualties	Regime: ED	−0.109	[−1.108; 0.896]	1:1

Note: The model includes 306 observations from 23 countries. Income (GDP per capita, PPP) was log-transformed for normalisation and then standardised for estimation. PB was standardised for estimation.

Influence of Burden on the Local Ecosystem Against Competing Explanations

This robustness test examines the effects of predictors on conflict outcomes while accounting for variables of competing explanations. Firstly, we included cultural diversity as an additional predictor to assess whether the observed relation between burden on the local ecosystem and conflict persists after controlling for cultural diversity.

The model structure is as follows:

$$\text{Income}_{i,j} = \beta_0 + \beta_1 \times \text{PB}_{i,j} + \beta_2 \times \text{Political regime}_{i,j} + u_j^{(1)} + \varepsilon_{i,j}^{(1)};$$

$$\begin{aligned}\text{Conflict}_{i,j} = \alpha_0 + \alpha_1 \times \text{PB}_{i,j} + \alpha_2 \times \text{Political regime}_{i,j} + \alpha_3 \times \text{Income}_{i,j} + \alpha_4 \\ \times \text{Cultural diversity}_{i,j} + u_j^{(2)} + \varepsilon_{i,j}^{(2)},\end{aligned}$$

where $\text{Income}_{i,j}$, $\text{PB}_{i,j}$, $\text{Political regime}_{i,j}$ and $\text{Cultural diversity}_{i,j}$ are the income level, PB, political regime and cultural diversity in country j in year i, respectively; $\text{Conflict}_{i,j}$ is the presence of conflict in country j in year i; $u_j^{(1)}$ and $u_j^{(2)}$ are the random effects of each country; and $\varepsilon_{ij}^{(1)}$ and $\varepsilon_{ij}^{(2)}$ are the error terms.

The same model specification is applied when using conflict intensity (measured by casualties) as the main dependent variable.

The data for cultural diversity was sourced from The Quality of Government (QoG) Institute.[95]

Presence of conflict models use a Bernoulli likelihood, while conflict-intensity models use a Student-t likelihood.

The results of this robustness test are reported in the following tables.

Table A11. Coefficient estimates for the conflict model with burden on the local ecosystem and cultural diversity as main predictors.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.788	[8.588; 8.988]	40000:1
Income	PB	0.127	[0.067; 0.187]	40000:1
Income	Regime: EA	0.231	[0.188; 0.275]	40000:1
Income	Regime: ED	0.337	[0.287; 0.386]	40000:1
Income	Regime: LD	0.446	[0.369; 0.522]	40000:1
Conflict	(Intercept)	−3.371	[−4.446; −2.42]	1:40000
Conflict	PB	1.387	[0.733; 2.089]	40000:1
Conflict	Income	−1.018	[−1.538; −0.509]	1:39999
Conflict	Cultural diversity	0.813	[0.019; 1.64]	44:1
Conflict	Regime: EA	−0.538	[−1.002; −0.072]	1:83
Conflict	Regime: ED	−1.486	[−2.099; −0.883]	1:40000
Conflict	Regime: LD	−3.499	[−6.226; −1.167]	1:1211

Note: The model includes 3963 observations from 146 countries. Income (GDP per capita, PPP) and PB were logtransformed for normalization and then standardized for estimation.

Table A12. Coefficient estimates for the casualties model with burden on the local ecosystem and cultural diversity as main predictors.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	7.924	[7.602; 8.251]	40000:1
Income	PB	0.157	[0.08; 0.235]	19999:1
Income	Regime: EA	0.342	[0.271; 0.413]	40000:1
Income	Regime: ED	0.200	[0.092; 0.307]	7999:1
Income	Regime: LD	2.619	[1.173; 4.058]	2856:1
Casualties	(Intercept)	5.940	[5.47; 6.403]	40000:1
Casualties	PB	0.467	[0.197; 0.741]	3332:1
Casualties	Income	−0.195	[−0.513; 0.114]	1:8
Casualties	Cultural diversity	0.071	[−0.291; 0.436]	2:1
Casualties	Regime: EA	−0.298	[−0.647; 0.05]	1:20
Casualties	Regime: ED	−0.245	[−0.723; 0.237]	1:5
Casualties	Regime: LD	−0.618	[−2.37; 1.167]	1:3

Note: The model includes 655 observations from 42 countries. Income (GDP per capita, PPP) was log-transformed for normalisation and then standardised for estimation. PB and cultural diversity were standardised for estimation.

In another test, we included immigration as an additional predictor to assess whether the observed relation between burden on the local ecosystem and conflict persists after controlling for immigration.

The model structure is as follows:

$$\text{Income}_{i,j} = \beta_0 + \beta_1 \times \text{PB}_{i,j} + \beta_2 \times \text{Political regime}_{i,j} + u_j^{(1)} + \varepsilon_{i,j}^{(1)};$$

$$\text{Conflict}_{i,j} = \alpha_0 + \alpha_1 \times \text{PB}_{i,j} + \alpha_2 \times \text{Political regime}_{i,j} + \alpha_3 \times \text{Income}_{i,j} + \alpha_4 \times \text{Immigration}_{i,j} + u_j^{(2)} + \varepsilon_{i,j}^{(2)},$$

where $\text{Income}_{i,j}$, $\text{PB}_{i,j}$, $\text{Political regime}_{i,j}$ and $\text{Immigration}_{i,j}$ are the income level, PB, political regime and immigration in country j in year i, respectively; $\text{Conflict}_{i,j}$ is the presence of conflict in country j in year i; $u_j^{(1)}$ and $u_j^{(2)}$ are the random effects of each country; and $\varepsilon_{ij}^{(1)}$ and $\varepsilon_{ij}^{(2)}$ are the error terms.

The data for immigration was sourced from Our World in Data.[96]

Presence of conflict models use a Bernoulli likelihood. Coefficient estimates for the casualties model are not reported due to insufficiency of data (*number of observations*: 45; *number of countries*: 9).

The results of this robustness test are reported in the following table.

Table A13. Coefficient estimates for the conflict model with burden on the local ecosystem and immigration as main predictors.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.633	[8.437; 8.828]	40000:1
Income	PB	0.038	[−0.084; 0.16]	3:1
Income	Regime: EA	0.277	[0.167; 0.387]	40000:1
Income	Regime: ED	0.438	[0.318; 0.559]	40000:1
Income	Regime: LD	0.829	[0.641; 1.017]	40000:1
Conflict	(Intercept)	−3.776	[−5.483; −2.333]	1:40000
Conflict	PB	0.850	[0.038; 1.743]	50:1
Conflict	Income	−1.407	[−2.46; −0.496]	1:1249
Conflict	Immigration	0.077	[−0.679; 0.901]	1:1
Conflict	Regime: EA	−0.055	[−1.193; 1.082]	1:1
Conflict	Regime: ED	−0.624	[−1.978; 0.689]	1:5
Conflict	Regime: LD	−2.257	[−5.698; 0.584]	1:14

Note: The model includes 735 observations from 147 countries. Income (GDP per capita, PPP), PB and immigration were log-transformed for normalisation and then standardised for estimation.

In the next test, we included inequality as an additional predictor to assess whether the observed relation between burden on the local ecosystem and conflict persists after controlling for inequality.

The model structure is as follows:

$$\text{Income}_{i,j} = \beta_0 + \beta_1 \times \text{PB}_{i,j} + u_j^{(1)} + \varepsilon_{i,j}^{(1)};$$

$$\text{Inequality}_{i,j} = \gamma_0 + \gamma_1 \times \text{Income}_{i,j} + u_j^{(2)} + \varepsilon_{i,j}^{(2)};$$

$$\text{Conflict}_{i,j} = \alpha_0 + \alpha_1 \times \text{PB}_{i,j} + \alpha_2 \times \text{Income}_{i,j} + \alpha_3 \times \text{Inequality}_{i,j} + u_j^{(3)} + \varepsilon_{i,j}^{(3)},$$

where $\text{Income}_{i,j}$, $\text{PB}_{i,j}$, Political $\text{regime}_{i,j}$ and $\text{Inequality}_{i,j}$ are the income level, PB, political regime and inequality in country j in year i, respectively; $\text{Conflict}_{i,j}$ is the presence of conflict in country j in year i; $u_j^{(1)}$ and $u_j^{(2)}$ are the random effects of each country; and $\varepsilon_{ij}^{(1)}$ and $\varepsilon_{ij}^{(2)}$ are the error terms.

To account for potential heterogeneity across political systems, the models were estimated separately for each regime type rather than including regime as a predictor within the equations. The same model specification is applied when using conflict intensity (measured by casualties) as the main dependent variable.

The measure of inequality used in this study – namely, the top 10 per cent income share – was sourced from the World Inequality Database.[97]

Presence of conflict models use a Bernoulli likelihood, while conflict-intensity models use a Student-t likelihood. Coefficient estimates for the casualties model in closed autocracies and liberal democracies are not reported due to insufficiency of data. The results of this robustness test are reported in the following tables.

Table A14. Coefficient estimates for the conflict model with burden on the local ecosystem and inequality as main predictors. Regime: Closed autocracies.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.441	[8.007; 8.866]	40000:1
Income	PB	−0.162	[−0.31; −0.016]	1:65
Inequality	(Intercept)	0.172	[0.157; 0.187]	40000:1
Inequality	Income	0.017	[0.011; 0.023]	40000:1
Conflict	(Intercept)	−6.977	[−13.606; −2.52]	1:689
Conflict	PB	4.584	[0.936; 11.071]	285:1
Conflict	Income	−5.432	[−9.364; −2.453]	1:40000
Conflict	Inequality	1.190	[−0.371; 3.121]	12:1

Note: The model includes 326 observations from 39 countries. Income (GDP per capita, PPP) and PB were log-transformed for normalization and then standardized for estimation. Inequality was standardized for estimation.

Table A15. Coefficient estimates for the conflict model with burden on the local ecosystem and inequality as main predictors. Regime: Electoral autocracies.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.479	[8.258; 8.699]	40000:1
Income	PB	0.200	[0.089; 0.311]	5713:1
Inequality	(Intercept)	0.166	[0.158; 0.175]	40000:1
Inequality	Income	−0.007	[−0.012; −0.002]	1:287
Conflict	(Intercept)	−2.572	[−3.682; −1.66]	1:40000
Conflict	PB	1.689	[0.793; 2.766]	19999:1
Conflict	Income	0.525	[−0.151; 1.248]	14:1
Conflict	Inequality	0.138	[−0.269; 0.545]	3:1

Note: The model includes 1087 observations from 84 countries. Income (GDP per capita, PPP) and PB were log-transformed for normalization and then standardized for estimation. Inequality was standardized for estimation.

Table A16. Coefficient estimates for the conflict model with burden on the local ecosystem and inequality as main predictors. Regime: Electoral democracies.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.854	[8.634; 9.071]	40000:1
Income	PB	0.313	[0.185; 0.441]	40000:1
Inequality	(Intercept)	0.161	[0.15; 0.172]	40000:1
Inequality	Income	−0.001	[−0.008; 0.005]	1:2
Conflict	(Intercept)	−7.542	[−10.914; −5.124]	1:40000
Conflict	PB	1.634	[−0.471; 3.805]	15:1
Conflict	Income	−1.102	[−2.557; 0.223]	1:18
Conflict	Inequality	−0.911	[−1.678; −0.217]	1:247

Note: The model includes 1067 observations from 81 countries. Income (GDP per capita, PPP) and PB were log-transformed for normalization and then standardized for estimation. Inequality was standardized for estimation.

Table A17. Coefficient estimates for the conflict model with burden on the local ecosystem and inequality as main predictors. Regime: Liberal democracies.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	10.261	[10.03; 10.497]	40000:1
Income	PB	0.404	[0.288; 0.521]	40000:1
Inequality	(Intercept)	0.133	[0.116; 0.15]	40000:1
Inequality	Income	0.018	[0.015; 0.021]	40000:1
Conflict	(Intercept)	−15.732	[−29.226; −8.842]	1:40000
Conflict	PB	1.406	[−2.541; 5.143]	4:1
Conflict	Income	0.858	[−2.3; 3.405]	3:1
Conflict	Inequality	3.983	[1.815; 7.022]	39999:1

Note: The model includes 968 observations from 43 countries. Income (GDP per capita, PPP) and PB were log-transformed for normalization and then standardized for estimation. Inequality was standardized for estimation.

Table A18. Coefficient estimates for the casualties model with burden on the local ecosystem and inequality as main predictors. Regime: Electoral autocracies.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.279	[7.894; 8.671]	40000:1
Income	PB	0.146	[0.021; 0.268]	84:1
Inequality	(Intercept)	0.177	[0.16; 0.193]	40000:1
Inequality	Income	−0.002	[−0.013; 0.009]	1:2
Casualties	(Intercept)	5.516	[5.031; 5.991]	40000:1
Casualties	PB	0.371	[−0.007; 0.743]	36:1
Casualties	Income	−0.086	[−0.534; 0.329]	1:2
Casualties	Inequality	−0.135	[−0.344; 0.08]	1:9

Note: The model includes 268 observations from 30 countries. Income (GDP per capita, PPP) was log-transformed for normalization and then standardized for estimation. PB and inequality were standardized for estimation.

Table A19. Coefficient estimates for the casualties model with burden on the local ecosystem and inequality as main predictors. Regime: Electoral democracies.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.699	[7.85; 9.566]	40000:1
Income	PB	1.071	[0.649; 1.478]	40000:1
Inequality	(Intercept)	0.166	[0.139; 0.192]	40000:1
Inequality	Income	0.010	[−0.003; 0.024]	14:1
Casualties	(Intercept)	5.734	[4.626; 6.871]	40000:1
Casualties	PB	0.689	[−0.378; 2.07]	8:1
Casualties	Income	−0.921	[−1.572; −0.302]	1:701
Casualties	Inequality	0.001	[−0.287; 0.288]	1:1

Note: The model includes 139 observations from 13 countries. Income (GDP per capita, PPP) was log-transformed for normalisation and then standardised for estimation. PB and inequality were standardised for estimation.

Lastly, we included property rights index as an additional predictor to assess whether the observed relation between burden on the local ecosystem and conflict persists after controlling for property rights index.

The model structure is as follows:

$$\text{Income}_{i,j} = \beta_0 + \beta_1 \times \text{PB}_{i,j} + \beta_2 \times \text{Political regime}_{i,j} + \beta_3 \times \text{Property rights}_{i,j} + u_j^{(1)} + \varepsilon_{i,j}^{(1)};$$

$$\text{Conflict}_{i,j} = \alpha_0 + \alpha_1 \times \text{PB}_{i,j} + \alpha_2 \times \text{Political regime}_{i,j} + \alpha_3 \times \text{Income}_{i,j} + \alpha_4 \times \text{Property rights}_{i,j} + u_j^{(2)} + \varepsilon_{i,j}^{(2)},$$

where $\text{Income}_{i,j}$, $\text{PB}_{i,j}$, $\text{Political regime}_{i,j}$ and $\text{Property rights}_{i,j}$ are the income level, PB, political regime and property rights index in country j in year i, respectively; $\text{Conflict}_{i,j}$ is the presence of conflict in country j in year i; $u_j^{(1)}$ and $u_j^{(2)}$ are the random effects of each country; and $\varepsilon_{ij}^{(1)}$ and $\varepsilon_{ij}^{(2)}$ are the error terms.

The same model specification is applied when using conflict intensity (measured by casualties) as the main dependent variable.

The data for property rights was sourced from the V-Dem research institute.[98]

Presence of conflict models use a Bernoulli likelihood, while conflict-intensity models use a Student-t likelihood.

The results of this robustness test are reported in the following tables.

Table A20. Coefficient estimates for the conflict model with burden on the local ecosystem and property rights as main predictors.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.849	[8.666; 9.026]	40000:1
Conflict	(Intercept)	−4.294	[−5.4; −3.278]	1:40000
Income	PB	0.166	[0.11; 0.224]	40000:1
Income	Property right	0.213	[0.183; 0.243]	40000:1
Income	Regime: EA	0.238	[0.196; 0.281]	40000:1
Income	Regime: ED	0.251	[0.201; 0.302]	40000:1
Income	Regime: LD	0.320	[0.246; 0.395]	40000:1
Conflict	PB	1.280	[0.61; 2.004]	9999:1
Conflict	Income	−0.812	[−1.344; −0.278]	1:740
Conflict	Property right	−1.129	[−1.548; −0.72]	1:40000
Conflict	Regime: EA	−0.348	[−0.825; 0.13]	1:12
Conflict	Regime: ED	−0.652	[−1.283; −0.012]	1:43
Conflict	Regime: LD	−2.638	[−5.51; −0.222]	1:64

Note: The model includes 4343 observations from 163 countries. Income (GDP per capita, PPP) and PB were logtransformed for normalization and then standardized for estimation. Property right was standardized for estimation.

Table A21. Coefficient estimates for the casualties model with burden on the local ecosystem and property rights as main predictors.

Outcome	Variable	Estimate	CI	BF
Income	(Intercept)	8.051	[7.774; 8.321]	40000:1
Income	PB	0.143	[0.068; 0.217]	39999:1
Income	Property right	0.313	[0.234; 0.392]	40000:1
Income	Regime: EA	0.267	[0.196; 0.338]	40000:1
Income	Regime: ED	0.060	[−0.046; 0.168]	6:1
Income	Regime: LD	2.077	[0.845; 3.317]	1537:1
Casualties	(Intercept)	5.908	[5.442; 6.372]	40000:1
Casualties	PB	0.474	[0.212; 0.746]	3999:1
Casualties	Income	−0.176	[−0.513; 0.149]	1:6
Casualties	Property right	−0.066	[−0.378; 0.245]	1:2
Casualties	Regime: EA	−0.275	[−0.624; 0.074]	1:15
Casualties	Regime: ED	−0.236	[−0.74; 0.269]	1:5
Casualties	Regime: LD	−0.581	[−2.363; 1.205]	1:3

Note: The model includes 667 observations from 44 countries. Income (GDP per capita, PPP) was log-transformed for normalisation and then standardised for estimation. PB and property right were standardised for estimation.

Bayesian Network Structure learning

This robustness test examines the relations among all predictors and conflict outcomes using Bayesian structure learning, which uses counterfactual reasoning to highlight the most likely causal paths among a set of variables.[99] We employed the *bnlearn* R package[100] to explore the probabilistic dependencies between variables and to assess whether the structure supports the hypothesised model specification.

For the conflict model, we applied the tabu search and hill-climbing (hc) algorithms, optimising the Bayesian Dirichlet equivalence (BDE) score. For the casualties model, we used the same algorithms, but optimised both the Bayesian Information Criterion (BIC) and the Akaike Information Criterion (AIC) scores.

The results of this Bayesian structure learning analysis are presented in the following figures.

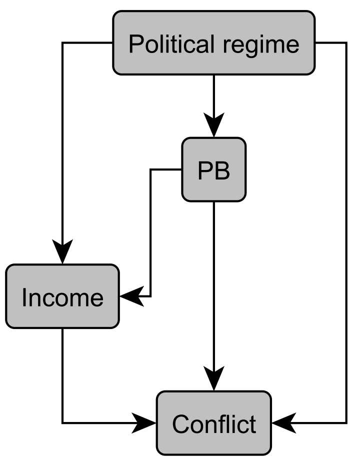

Figure A1. Directed Acyclic Graph (DAG) structure learned by the tabu search tabu and hill-climbing (hc) algorithms optimising the Bayesian Dirichlet equivalence (BDE) score.

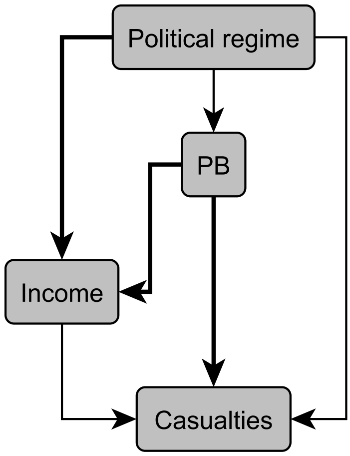

Figure A2. DAG structure learned by the tabu and hc algorithms optimising BIC and AIC scores. All arcs are present in the DAG learned by optimising the AIC score. Only bold arcs are present in the DAG learned by optimising the BIC score.